

Analysis of Urban Growth Dynamics in the United Arab Emirates

Using Multi-Temporal Sentinel-2 Satellite Imagery
(2018-2024)

A Capstone Project Report



Authors:

Yessmine Chaabouni

Sahar Feki

Supervisor: Ali Ben Abbes

Institution: Manouba School of Engineering

Date: September-December 2025

Abstract

Urbanization is one of the most significant forms of land use change in arid regions, particularly in rapidly developing countries such as the United Arab Emirates (UAE). Accurate and up-to-date monitoring of urban expansion is essential for sustainable planning, environmental protection, and infrastructure management. This project proposes an automated and scalable framework for mapping and quantifying **urban growth** in the UAE between 2018 and 2024 using **Sentinel-2** satellite imagery and the **Google Earth Engine** (GEE) cloud computing platform.

The methodology relies on the generation of annual cloud-free composites, the extraction of spectral indices (NDVI, NDBI, MNDWI, BUI, and a desert-adapted urban index), and supervised **land cover classification** using a **Random Forest** algorithm. Post-classification refinement and robust area computation strategies are applied to ensure reliable results over large spatial extents. The results show a significant increase in urban area over the study period, with urban surfaces expanding from approximately 7,200 km² in 2018 to more than 11,500 km² in 2024, representing an overall growth of more than 60%.

This work demonstrates the effectiveness of cloud-based geospatial platforms combined with machine learning techniques for large-scale, long-term urban monitoring in desert environments and provides a reproducible framework for future urban studies in the region.

Acknowledgments

We would like to express our sincere gratitude to our supervisor Mr. Ali Ben Abbes for his guidance, valuable advice, and continuous support throughout this project. We also thank our professors and the entire teaching staff of Manouba School of Engineering for providing us with the scientific and technical background necessary to complete this work. Finally, we thank our families and friends for their encouragement and support during the realization of this project.

Yessmine Chaabouni

Sahar Feki

Date: January 15, 2026

Table of Contents

Chapter 1	Introduction	1
1.1	Problem Statement and Motivation	1
1.2	Project Objectives	2
1.2.1	Primary Objective	2
1.2.2	Secondary Objectives	2
1.3	Project Scope and Limitations	4
1.3.1	Scope	4
1.3.2	Limitations	4
1.4	Report Structure	5
1.5	Study Area Description	6
1.5.1	Geographical and Climatic Context	6
1.5.2	Urban Context	6
1.5.3	Temporal Significance	6
Chapter 2	Literature Review	7
2.1	Urban Remote Sensing and Land Cover Mapping	7
2.2	Spectral Indices for Urban and Land Cover Discrimination	7
2.3	Classification Algorithms: Random Forest vs. SVM and Others	8
2.4	Cloud Computing Platforms and Google Earth Engine	9
2.5	Previous Studies in Similar Regions	10
2.6	Identified Research Gaps and Project Contribution	11

Chapter 3 Methodology and System Design	13
3.1 Overall Research Framework	13
3.2 Phase 1: AOI definition - Study Area and Data Sources	15
3.2.1 Study Area Definition	15
3.2.2 Temporal Scope and Data Selection	15
3.2.3 Data Specifications	15
3.3 Data Preprocessing and Compositing	16
3.3.1 Cloud Masking Implementation	16
3.3.2 Annual Composite Generation	16
3.4 Phase 3: Spectral Index Computation	17
3.4.1 Spectral Index Choice Selection	17
3.4.2 Standard Indices	18
3.4.3 Custom Desert Urban Index (DUI)	19
3.5 Phase 4: Training Data and Classification	19
3.5.1 Training Data Preparation	19
3.6 Phase 5: Supervised Classification: Random Forest Classifier Configuration	23
3.6.1 Overview	23
3.6.2 Image Classification Using Random Forest	23
3.6.3 Post-Processing	27
3.7 Phase 6: Rule Based Corrections - Post-Classification Analysis	28
3.7.1 Overview	28
3.7.2 Spatial Smoothing (Post-Classification Filtering)	29
3.7.3 Desert-Optimized Classification	29
3.8 Phase 7: Urban Area Quantification	30
3.8.1 Change Detection Methodology	32

3.9 Phase 8: Dashboard Development	32
3.9.1 Interactive Interface Design	32
3.9.2 Dashboard Features	32
3.9.3 Data Export and Reproducibility	33
Chapter 4 Accuracy, Results, Analysis, and Discussion	36
4.1 Accuracy Assessment Protocol	36
4.1.1 Purpose of Accuracy Assessment	36
4.1.2 Reference Data for Validation	36
4.1.3 Validation Procedure	37
4.1.4 Accuracy Metrics	37
4.1.5 Results Interpretation	38
4.1.6 Limitations and Considerations	39
4.2 Data and Spectral Index Results	40
4.2.1 Spectral Index Analysis and Interpretation	41
4.3 Classification Results and Classification Accuracy Assessment	48
4.3.1 Classification Results	48
4.3.2 Classification Accuracy Assessment	48
4.4 Quantitative Urban Growth Analysis (2018-2024)	49
4.4.1 Statistics	49
4.4.2 Spatial Patterns of Urban Expansion	50
4.4.3 Discussion	51
4.5 Dashboard Functionality and Operational Results	52
4.6 Challenges and Methodological Solutions	53
4.7 Discussion	53

Chapter 5	Project Management Methodology	55
5.1	Agile-Scrum Framework	55
5.2	Epic Breakdown and Implementation	55
5.2.1	EPIC 1: Data Acquisition and Preprocessing	56
5.2.2	EPIC 2: Feature Engineering	57
5.2.3	EPIC 3: Training Dataset Construction	58
5.2.4	EPIC 4: Supervised Classification	59
5.2.5	EPIC 5: Accuracy Assessment	60
5.2.6	EPIC 6: Urban Area Statistics and Change Detection	61
5.2.7	EPIC 7: Visualization and Dashboard	62
5.2.8	EPIC 8: Export and Deliverables	63
5.3	Project Release Planning and Execution	63
5.3.1	Release 1: Data and Classification Foundation	63
5.3.2	Release 2: Validation, Correction, and Urban Growth Analytics	64
5.3.3	Release 3: Dashboard, Visualization, and Final Delivery	65
5.4	Release Management Strategy	65
5.5	Project Management Outcomes	66
Chapter 6	Conclusion and Future Work	67
6.1	Overall Project Summary and Achievements	67
6.2	Answer to the Initial Problem Statement	68
6.3	Critical Discussion of Project Limitations	68
6.4	Future Work and Perspectives	69

7	References	71
	Appendix A: Google Earth Engine Code	73
	A Main Processing Script	73
	B Classification Function	73
	Appendix B: Detailed Accuracy Assessment Tables	74
	Appendix C: Training and Validation Sample Locations	74

List of Figures

1	Conceptual workflow illustrating the relationship between project objectives and methodological steps	3
2	Specific workflow implemented in Google Earth Engine JavaScript API . . .	14
3	Evolution of training polygon distribution for 2018 land cover classification .	20
4	Interactive dashboard with year selector, layer toggles, and statistics panel .	32

List of Tables

1	Comparison of Classification Algorithms for Urban Mapping	9
2	Representative Urban Growth Studies in Arid Regions	11
3	Specific Data Sources and Specifications	15
4	Comprehensive Suite of Spectral Indices	18
5	Spectral Indices Used in Classification for Land Cover Discrimination	19
6	Training Dataset Composition (2018)	20
7	Input Features for Random Forest Classification	26

Chapter 1

1 Introduction

1.1 Problem Statement and Motivation

Urban growth is one of the most visible and impactful forms of land surface transformation, particularly in arid and semi-arid environments. The United Arab Emirates (UAE) has experienced extremely rapid urbanization over the past decades, driven by population growth, economic development, and large-scale infrastructure projects. This rapid expansion exerts strong pressure on natural resources, ecosystems, and land availability, making accurate monitoring of urban dynamics a critical requirement for sustainable planning and decision-making.

Traditional urban mapping methods based on field surveys or manual digitization are costly, time-consuming, and difficult to update regularly over large areas. Remote sensing offers an efficient alternative by providing consistent, repetitive, and large-scale observations of the Earth's surface. However, urban mapping in desert environments remains challenging due to the spectral similarity between built-up areas and bare soil or rocky surfaces, which often leads to misclassification.

At the same time, the increasing availability of high-resolution satellite data such as Sentinel-2 and the emergence of cloud computing platforms such as Google Earth Engine (GEE) have created new opportunities for large-scale, multi-temporal urban analysis. This project is motivated by the need to exploit these modern tools to produce a reliable, reproducible, and automated system for monitoring urban growth in the UAE.

1.2 Project Objectives

1.2.1 Primary Objective

The main objective of this project is to develop an automated and robust framework for mapping and quantifying urban growth in the United Arab Emirates between 2018 and 2024 using Sentinel-2 imagery and Google Earth Engine. This framework aims to provide a scalable methodology for monitoring urbanization in arid environments with high accuracy and computational efficiency.

1.2.2 Secondary Objectives

- SO1: Data Processing and Compositing:** To generate annual cloud-free composite images from Sentinel-2 time series using advanced preprocessing techniques including cloud masking, atmospheric correction, and temporal compositing optimized for arid regions.
- SO2: Spectral Analysis:** To extract and analyze relevant spectral indices (NDVI, NDBI, MNDWI, BUI, and desert-adapted urban indices) for enhanced discrimination between urban areas, vegetation, bare soil, and water bodies in the UAE's desert environment.
- SO3: Machine Learning Classification:** To implement and train a supervised machine learning classifier (Random Forest) for accurate land cover mapping, leveraging the comprehensive spectral information from Sentinel-2 imagery and derived indices.
- SO4: Classification Enhancement:** To improve classification reliability in desert environments through rule-based corrections (particularly for vegetation-desert confusion) and systematic post-processing techniques including spatial smoothing and accuracy assessment.

- SO5: Quantitative Analysis:** To compute accurate urban area statistics and analyze temporal trends of urban expansion across the UAE, including growth rates, spatial patterns, and comparative analysis between different emirates and time periods.
- SO6: Visualization and Dissemination:** To produce cartographic outputs, change detection maps, and an interactive visualization dashboard for result interpretation, facilitating stakeholder engagement and supporting evidence-based urban planning decisions.
- SO7: Methodological Validation:** To validate the developed framework through rigorous accuracy assessment, comparison with existing datasets, and demonstration of reproducibility for application in other arid and semi-arid regions.

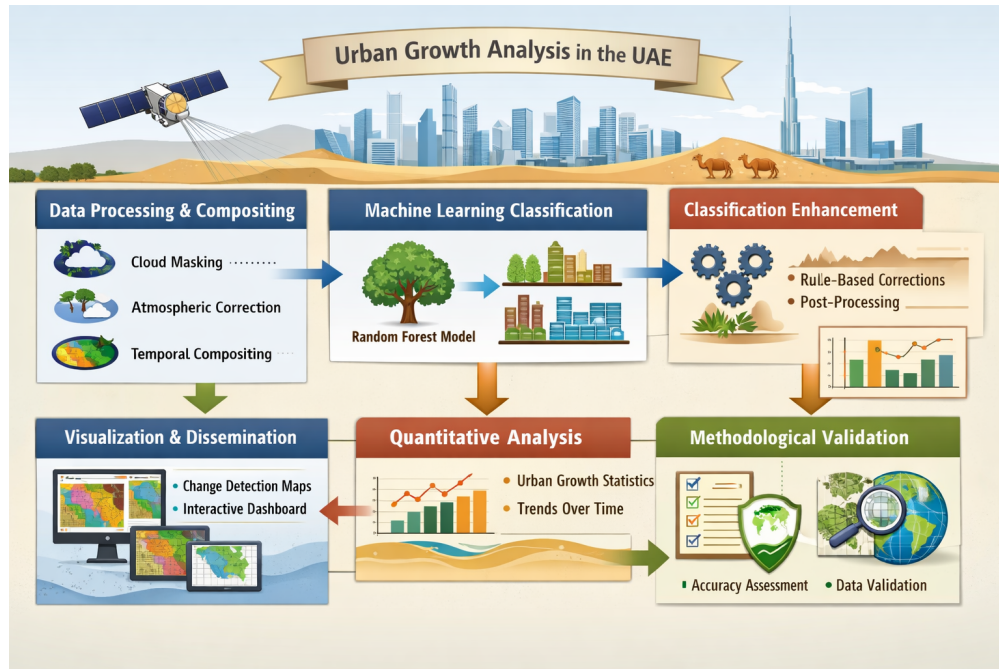


Figure 1: Conceptual workflow illustrating the relationship between project objectives and methodological steps

Alignment with Research Goals

The defined objectives systematically address the research needs for **Spatio-temporal analysis** that tracks urban expansion patterns over a 7-year period, **Methodological innovation** which is developing desert-adapted remote sensing techniques, **Practical application** where the creation of tools and outputs that will be useful for urban planners and policymakers and **Scalability** which is establishing a framework applicable to other rapidly urbanizing arid regions.

These objectives collectively ensure that the project delivers both scientific insights into UAE’s urbanization dynamics and practical tools for sustainable urban development monitoring.

1.3 Project Scope and Limitations

1.3.1 Scope

This project encompasses the spatial analysis of the entire territory of the United Arab Emirates through annual monitoring from 2018 to 2024. The research utilizes Sentinel-2 Level-2A surface reflectance data as its primary Earth observation source, processed entirely within the Google Earth Engine cloud computing platform. The methodological framework employs a four-class land cover classification system distinguishing Urban, Vegetation, Bare Soil/Desert, and Water categories. The analytical approach focuses on pixel-based supervised classification enhanced through multiple spectral indices to improve thematic discrimination in arid environments.

1.3.2 Limitations

The study acknowledges several methodological constraints inherent to remote sensing analysis in arid environments. The spatial resolution of Sentinel-2 data (10–20 m) imposes limitations on detecting very small urban features and fine-scale built structures. Persis-

tent spectral confusion remains between bare soil, rocky surfaces, and built-up materials, particularly in mountainous regions where reflectance characteristics overlap significantly. The classification accuracy depends substantially on manually digitized training data, which may not fully capture the complete landscape variability across the UAE's diverse terrain. The analysis focuses primarily on horizontal urban expansion patterns rather than vertical densification and three-dimensional urban growth. Temporal granularity is limited to annual composites, potentially missing intra-annual seasonal variations in land cover. Despite advanced compositing techniques, residual atmospheric effects and cloud cover may persist in certain areas, particularly in coastal regions with higher humidity.

1.4 Report Structure

This report is organized into five comprehensive chapters:

Chapter 1: Introduction: Presents the project context, problem statement, objectives, scope, and study area description.

Chapter 2: Literature Review: Surveys existing work in urban remote sensing, classification methods, spectral indices, and cloud computing platforms.

Chapter 3: Methodology: Details the proposed framework, system design, and implementation steps including data preprocessing, classification, and validation.

Chapter 4: Results and Analysis: Presents empirical findings, accuracy assessment metrics, urban growth statistics, and spatial patterns.

Chapter 5: Project Management Methodology: The project was managed using an Agile methodology inspired by the Scrum framework, adapted for scientific research workflows

Chapter 6: Conclusion and Future Work: Summarizes key contributions, discusses limitations, and proposes directions for subsequent research.

1.5 Study Area Description

1.5.1 Geographical and Climatic Context

The study area encompasses the entire territory of the United Arab Emirates (UAE), located in the southeastern Arabian Peninsula between latitudes 22.5°N and 26.5°N and longitudes 51.5°E and 56.5°E. The UAE's geographical characteristics include:

1.5.2 Urban Context

The United Arab Emirates represents a compelling case study for urban growth analysis due to its status as one of the world's fastest urbanizing regions, with unprecedented growth rates since the 1970s. The country features major metropolitan centers including Abu Dhabi (the capital), Dubai (a global commercial hub), Sharjah, and Al Ain (an inland oasis city). This period witnessed extensive infrastructure development through mega-projects such as artificial islands, iconic skyscrapers, and comprehensive transportation networks. These rapid transformations present significant planning challenges, particularly in balancing accelerated development with environmental sustainability in a water-scarce region.

1.5.3 Temporal Significance

The selected temporal window from 2018 to 2024 corresponds to several key developments in the UAE. This period encompasses the post-Expo 2020 infrastructure expansion phase following Dubai's hosting of the global event. It captures the continuous urbanization process transforming desert landscapes into urban environments at an unprecedented scale. The timeframe aligns perfectly with the complete operational availability of the Sentinel-2 satellite constellation, ensuring consistent and comprehensive Earth observation data coverage. Additionally, this period reflects increased focus on sustainable urban development and climate initiatives within the Gulf region, providing valuable insights into contemporary urban growth patterns during a phase of heightened environmental awareness.

Chapter 2

2 Literature Review

2.1 Urban Remote Sensing and Land Cover Mapping

Remote sensing has long been recognized as an essential tool for land use and land cover (LULC) mapping due to its ability to provide consistent, repetitive, and large-scale observations. Early urban studies relied mainly on medium-resolution sensors such as Landsat, while recent years have seen a shift toward **higher-resolution sensors such as Sentinel-2, which offers improved spatial, spectral, and temporal resolution.**

Urban environments are particularly challenging to classify because they are spectrally heterogeneous and often confused with bare soil, especially in arid regions. As a result, **modern urban mapping approaches increasingly rely on a combination of spectral indices, multi-temporal analysis, and machine learning techniques.**

2.2 Spectral Indices for Urban and Land Cover Discrimination

Several spectral indices have been proposed to enhance specific land cover types:

-**Normalized Difference Vegetation Index (NDVI):** Widely adopted for vegetation detection and vigor assessment, calculated as $\frac{NIR-Red}{NIR+Red}$.

-**Modified Normalized Difference Water Index (MNDWI):** Specifically designed to enhance water bodies while suppressing signals from built-up areas and vegetation.

-**Normalized Difference Built-up Index (NDBI):** Commonly employed to highlight urban and built-up surfaces by exploiting the spectral contrast between SWIR and NIR bands.

-**Built-up Index (BUI):** Combines NDVI and NDBI ($BUI = NDBI - NDVI$) to improve urban detection by reducing confusion with vegetation.

However, in desert environments, NDBI often responds strongly to bare soil and rocks. Therefore, recent studies suggest combining multiple indices and introducing customized indices or rule-based corrections, as done in this project through a desert-adapted urban index.

2.3 Classification Algorithms: Random Forest vs. SVM and Others

Traditional pixel-based classifiers such as Maximum Likelihood Classification (MLC) have been progressively replaced by machine learning algorithms that offer superior performance in handling complex, non-linear spectral relationships. Among the prominent machine learning approaches:

- Support Vector Machines (SVM)**: Known for strong performance with limited training data through effective margin maximization in high-dimensional feature spaces. However, SVM performance is highly sensitive to parameter tuning (kernel selection, regularization parameter C , γ) and may struggle with class imbalance.

- Random Forest (RF)**: An ensemble learning method that constructs multiple decision trees during training and outputs the mode of their predictions. RF offers several advantages including robustness to noise, automatic feature importance estimation, resistance to overfitting, and minimal parameter tuning requirements.

- Other Algorithms**: Additional methods include Artificial Neural Networks (ANN), Decision Trees (CART), and Gradient Boosting Machines, each with specific strengths and computational requirements.

Comparative studies in remote sensing literature consistently demonstrate **Random Forest’s superiority for land cover mapping applications**. **Zhang2018** reported RF accuracy improvements of 5-15% over SVM for Sentinel-2 urban classification, while **Rodriguez2019** found RF particularly effective for heterogeneous landscapes. These performance advantages, combined with RF’s computational efficiency and interpretability, motivated its selection as the primary classifier for this project.

Table 1: Comparison of Classification Algorithms for Urban Mapping

Algorithm		Strengths	Limitations
Maximum Likelihood		Simple implementation, probabilistic outputs	Assumes normal distribution, sensitive to training data
Support Vector Machine	Vector	Effective with limited samples, handles high dimensions	Sensitive to parameters, computationally intensive
Random Forest		Robust to noise, handles non-linearity, feature importance	Memory intensive with many trees, potential overfitting
Neural Networks		Captures complex patterns, end-to-end learning	Requires large datasets, "black box" interpretation

2.4 Cloud Computing Platforms and Google Earth Engine

The exponential growth of satellite data volume presents significant challenges for local processing, including storage requirements, computational limitations, and processing time. Cloud computing platforms have emerged as essential tools for large-scale remote sensing analysis:

- Google Earth Engine (GEE)**: Provides access to petabytes of satellite imagery (including the complete Sentinel-2 archive) with parallel processing capabilities and a JavaScript/Python API for algorithm development.
- Amazon Web Services (AWS)** and **Microsoft Azure**: Offer cloud infrastructure for geospatial processing but require more setup and management compared to GEE’s purpose-built environment.
- Open Data Cube** and **OpenEO**: Open-source initiatives for cloud-based Earth observation processing with multi-provider support.

GEE has become particularly prominent in recent remote sensing research. **Gorelick2017** documented GEE’s transformative impact on large-scale environmental monitoring, while **Kumar2021** demonstrated its effectiveness for national-scale land cover mapping. For this project, GEE’s integrated data catalog, distributed computing infrastructure, and elimination of local storage constraints make it ideally suited for multi-temporal analysis across the UAE’s extensive territory.

2.5 Previous Studies in Similar Regions

Urban growth in arid and semi-arid regions has received increasing research attention, with several notable studies in Middle Eastern contexts:

- Gulf Region Studies**: Research in Saudi Arabia (**Alqurashi2016**), Qatar (**Finseca2020**), and Oman (**Charabi2021**) consistently report rapid urban expansion, particularly in coastal zones, with annual growth rates exceeding 5% %.
- Spectral Challenges**: These studies uniformly highlight the difficulty of distinguishing urban areas from bare soil and rock surfaces using conventional spectral indices, necessitating specialized approaches.
- Methodological Trends**: Recent works emphasize integrated approaches combining

multi-temporal analysis, machine learning classification, and cloud computing platforms for scalable processing.

Table 2: Representative Urban Growth Studies in Arid Regions

Study/Region	Time Period	Key Findings
Dubai, UAE (Hereher2019)	2000-2017	250% urban growth, primarily coastal expansion
Riyadh, Saudi Arabia (Alqurashi2016)	1990-2013	Complex urban patterns, desert-urban interface challenges
Doha, Qatar (Fonseca2020)	1987-2018	Rapid infill development, spectral confusion with desert

2.6 Identified Research Gaps and Project Contribution

Despite methodological advances in urban remote sensing, several significant gaps persist in existing literature:

Current Limitations in Literature

1. **Temporal Coverage:** Many studies analyze short time spans (2-3 years) rather than continuous multi-decadal trends
2. **Spatial Resolution:** Reliance on Landsat (30m) rather than higher-resolution Sentinel-2 (10m) for urban feature discrimination
3. **Processing Scalability:** Dependence on manual intervention rather than fully automated, reproducible workflows
4. **Environmental Adaptation:** Insufficient attention to desert-specific classification challenges and solutions

Project Contributions

This project addresses these gaps through several methodological innovations:

Contribution 1: Extended Temporal Analysis: Implementation of a seven-year continuous time series (2018-2024) using consistent methodology

Contribution 2: High-Resolution Processing: Utilization of Sentinel-2's 10m spatial resolution for improved urban feature detection

Contribution 3: Fully Automated Workflow: Development of a complete GEE-based pipeline from data preprocessing to change detection

Contribution 4: Desert-Adapted Methodology: Integration of specialized spectral indices and rule-based corrections for arid environments

Contribution 5: Comprehensive Outputs: Generation of both quantitative statistics and high-quality cartographic products for urban planning applications

This integrated approach represents a significant advancement in urban growth monitoring for arid regions, providing a scalable framework that balances methodological rigor with practical applicability for sustainable urban planning in rapidly developing desert environments.

Chapter 3

3 Methodology and System Design

3.1 Overall Research Framework

This project implements a fully automated, cloud-based workflow for urban growth analysis developed entirely within Google Earth Engine. The framework transforms raw Sentinel-2 imagery into quantified urban growth statistics through eight integrated phases (Figure 2):

Phase 1: AOI Definition: Definition of UAE boundaries using USDOS/LSIB dataset and spatial subsetting

Phase 2: Data Preprocessing: Cloud masking using SCL layer and annual composite generation (Oct-Apr)

Phase 3: Spectral Enhancement: Computation of NDVI, NDBI, MNDWI, BUI, and custom Desert Urban Index (DUI)

Phase 4: Training Data Generation: Manual digitization of 2018 reference polygons with 70-30 train-validation split

Phase 5: Random Forest Classification: Implementation with 100 trees using 10 spectral bands/indices as features

Phase 6: Rule-Based Correction: Desert-optimized post-processing with water and vegetation thresholds

Phase 7: Area Calculation: Robust urban area computation at 100m scale for memory efficiency

Phase 8: Dashboard Visualization: Interactive GEE UI with year selector, layer toggles, and statistics display



Figure 2: Specific workflow implemented in Google Earth Engine JavaScript API

3.2 Phase 1: AOI definition

Study Area and Data Sources

3.2.1 Study Area Definition

The UAE study area was precisely defined using the U.S. Department of State’s Large Scale International Boundary dataset:

```
1 var countries = ee.FeatureCollection("USDOS/LSIB_SIMPLE/2017");
2 var uae = countries.filter(ee.Filter.eq('country_na', 'United Arab
   Emirates'));
3 var aoi = uae.geometry();
4 Map.centerObject(aoi, 7);
```

Listing 1: AOI definition in GEE

3.2.2 Temporal Scope and Data Selection

The analysis spans 2018-2024 with dry season composites (October to April) to minimize atmospheric interference. Each annual composite integrates all available cloud-free Sentinel-2 scenes within this period using median reduction.

3.2.3 Data Specifications

Table 3: Specific Data Sources and Specifications

Data Type	Source/ID	Specifications
Sentinel-2 SR	COPERNICUS/S2_SR	Level-2A, 10-20m resolution
Boundary Data	USDOS/LSIB_SIMPLE/2017	UAE administrative boundaries
Cloud Masking	SCL band	Pixel-level classification
Training Data	Manual digitization	2018 reference year

3.3 Data Preprocessing and Compositing

3.3.1 Cloud Masking Implementation

Annual dry-season composites were generated using a median temporal reduction approach applied to Sentinel-2 SR imagery from October to April each year. This period was strategically selected to:

- Minimize cloud contamination (typically less than 5% in UAE dry season)
- Reduce seasonal vegetation variability
- Enhance stability of urban and desert spectral signatures

The specific cloud masking algorithm removes pixels classified as cloud shadow (Scene Classification Layer SCL=3), **medium cloud (8), high cloud (9), and cirrus (10)**:

```
1 function maskS2SR(img) {  
2   var scl = img.select('SCL');  
3   return img.updateMask(  
4     scl.neq(3).and(scl.neq(8)).and(scl.neq(9)).and(scl.neq(10))  
5   );  
6 }
```

Listing 2: Exact cloud masking function

3.3.2 Annual Composite Generation

For each year, the composite generation follows a specific temporal window and band selection strategy:

```

1 function annualComposite(year) {
2   var start = ee.Date.fromYMD(year, 10, 1);
3   var end = ee.Date.fromYMD(year + 1, 4, 30);
4
5   var collection = ee.ImageCollection('COPERNICUS/S2_SR')
6     .filterBounds(aoi)
7     .filterDate(start, end)
8     .map(maskS2SR);
9
10  var filtered = collection.map(function(img) {
11    return img.select(['B2', 'B3', 'B4', 'B8', 'B11', 'SCL'])
12      .copyProperties(img, ['system:time_start']);
13  });
14
15  var img = filtered.median().clip(aoi).toFloat();
16  return addIndices(img).set('year', year);
17 }

```

Listing 3: Annual composite function

3.4 Phase 3: Spectral Index Computation

3.4.1 Spectral Index Choice Selection

Spectral Index Notation

Throughout this study, spectral indices are referenced using standardized abbreviations: Normalized Difference Built-up Index (NDBI), Normalized Difference Vegetation Index (NDVI), Modified Normalized Difference Water Index (MNDWI), Index-Based Built-up Index (IBI), Enhanced Vegetation Index (EVI), Normalized Difference Snow Index (NDSI), Burn Area Index (BAI), and Soil Adjusted Vegetation Index (SAVI). These indices were computed from Sentinel-2 surface reflectance bands where B2, B3, B4, B8, and B11 correspond to blue (490

nm), green (560 nm), red (665 nm), near-infrared (842 nm), and shortwave infrared (1610 nm) wavelengths, respectively.

Table 4: Comprehensive Suite of Spectral Indices

Category	Index	Formula	Primary Application
3*Urban	NDBI	$\frac{B11-B8}{B11+B8}$	Built-up area indicator
	MNDWI	$\frac{B3-B11}{B3+B11}$	Water body extraction
	IBI	$\frac{NDBI - \frac{NDVI+MNDWI}{2}}{NDBI + \frac{NDVI+MNDWI}{2}}$	Enhanced urban feature extraction
2*Vegetation	NDVI	$\frac{B8-B4}{B8+B4}$	Vegetation vigor assessment
	EVI	$2.5 \times \frac{B8-B4}{B8+6B4-7.5B2+1}$	Atmospheric-resistant vegetation
4*Supplementary	SAVI	$\frac{B8-B4}{B8+B4+0.5} \times 1.5$	Soil-adjusted vegetation
	BAI	$\frac{1}{(0.1-B4)^2 + (0.06-B8)^2}$	Bare soil detection
	NDSI	$\frac{B3-B11}{B3+B11}$	Limited application
	SWIR Ratio	$B11/B8$	Desert vs. built-up discrimination

Index Selection Rationale

Following established UAE remote sensing literature, the combination of **NDVI**, **NDBI**, and **BUI** provides optimal discrimination between desert sands and built-up areas. The BUI index specifically addresses spectral confusion common in arid environments where bright desert surfaces often mimic urban spectral signatures.

3.4.2 Standard Indices

Four key spectral indices were computed to enhance specific land cover properties:

The study computes five spectral indices tailored for desert environments:

```
1 function addIndices(img) {
```

Table 5: Spectral Indices Used in Classification for Land Cover Discrimination

Index	Formula	Spectral Bands	Primary Application
NDVI	$\frac{NIR-Red}{NIR+Red}$	B8, B4	Vegetation vigor
NDBI	$\frac{SWIR-NIR}{SWIR+NIR}$	B11, B8	Built-up areas
BUI	$NDBI - NDVI$	Derived	Urban vs. desert discrimination
MNDWI	$\frac{Green-SWIR}{Green+SWIR}$	B3, B11	Water body extraction
DUI	$\times NDBI - NDVI - 0.5 \times MNDWI$	B9, B10, B11	Desert-adapted urban detection

```

2  var ndvi = img.normalizedDifference(['B8','B4']).rename('NDVI');
3  var ndbi = img.normalizedDifference(['B11','B8']).rename('NDBI');
4  var mndwi = img.normalizedDifference(['B3','B11']).rename('MNDWI');
5  var bui = ndbi.subtract(ndvi).rename('BUI');
6  var dui = ndbi.multiply(1.5).subtract(ndvi)
7           .add(mndwi.multiply(-0.5)).rename('DUI');
8  return img.addBands([ndvi, ndbi, mndwi, bui, dui]);
9 }

```

Listing 4: Spectral index computation

3.4.3 Custom Desert Urban Index (DUI)

The DUI represents an innovation for this project, combining NDBI, NDVI, and MNDWI with optimized weights for desert environments:

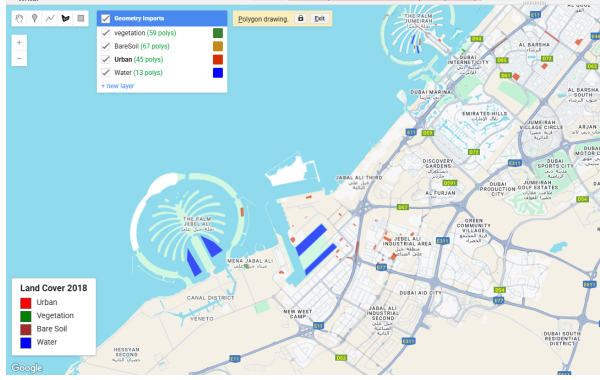
$$DUI = 1.5 \times NDBI - NDVI - 0.5 \times MNDWI$$

3.5 Phase 4: Training Data and Classification

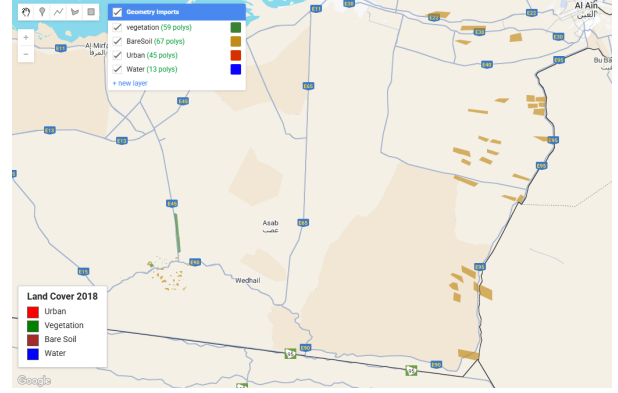
3.5.1 Training Data Preparation

Polygons Creation Training data were generated through meticulous manual digitization of reference polygons representing the four target land cover classes. For the baseline year 2018, a total of **184 polygons** were created: **45 urban**, **59 vegetation**, **67 bare soil**, and **13**

water polygons. Each polygon was digitized in homogeneous areas to ensure spectral purity and minimize within-class variability.



(a) Polygons Displayed in 2018 1.0



(b) Polygons Displayed in 2018 2.0

Figure 3: Evolution of training polygon distribution for 2018 land cover classification

Training data were generated exclusively from the 2018 composite to ensure temporal consistency. The dataset includes:

Table 6: Training Dataset Composition (2018)

Class	Label	Samples
Urban	0	800
Vegetation	1	600
Bare Soil	2	1000
Water	3	400
Total		2800

```

1 var trainingBands = ['B2','B3','B4','B8','B11','NDVI','NDBI','MNDWI','BUI',
2   ',DUI'];
3
4 function sampleTraining(polygon, classId, className, nSamples) {
5   return img2018.select(trainingBands)
6     .sample({
7       region: polygon,
8       scale: 30,
9       numPixels: nSamples,

```

```

9      seed: 42,
10     tileSize: 4,
11     geometries: false
12   })
13   .map(function(f) {
14     return f.set({
15       'class': classId,
16       'label': className
17     });
18   });
19 }

```

Listing 5: Training data sampling

Manual Digitization Protocol Training polygons were manually digitized through visual interpretation of RGB composites (B4, B3, B2) and cross-referenced with spectral index responses:

Urban: High NDBI (>0.1) and BUI (>0.1) with low NDVI (<0.1)

Vegetation: High NDVI (>0.2) and negative BUI values

Water: High MNDWI (>0.2) with near-zero NDVI

Bare Soil: Near-zero NDVI (≈ 0) and neutral BUI (≈ 0)

Training polygons were manually digitized by visually interpreting the RGB composite and cross-validating spectral index responses. Urban areas were characterized by high NDBI and BUI values with near-zero NDVI, vegetation exhibited high NDVI and negative BUI values, bare soil showed near-zero NDVI and neutral BUI, while water bodies were clearly identified using strong positive MNDWI values. This multi-layer verification ensured high-purity training samples.

Pixel-Based Sampling Strategy

Purpose

The objective of the sampling step is to create a reliable and representative set of reference samples that link spectral information extracted from satellite imagery to known land-cover classes. These samples are essential for training a supervised classification algorithm and for ensuring accurate discrimination between urban and non-urban land-cover types.

In this study, training samples were derived from manually digitized reference polygons and used to train a Random Forest classifier.

Sampling Approach

Rather than using entire polygons as single samples, a pixel-based sampling approach was adopted to allow the classifier to learn intra-class spectral variability. For each land-cover class:

A fixed number of pixels were randomly sampled from each polygon

Sampling was performed at the native 10 m spatial resolution of Sentinel-2

Samples were extracted exclusively from the 2018 annual composite

3.6 Phase 5: Supervised Classification:

Random Forest Classifier Configuration

3.6.1 Overview

The classification phase employs an AI-based approach for land cover mapping and urban growth detection. **The AI model uses a Random Forest classifier trained on spectral bands and indices from Sentinel-2 images to automatically classify land cover and detect urban growth by identifying patterns associated with built-up areas.** This machine learning approach enables accurate, repeatable classification across multiple time periods.

3.6.2 Image Classification Using Random Forest

Purpose of Classification After the sampling stage where representative training samples were extracted for each land cover class, the objective of classification is to assign a land cover class label to every pixel in the satellite image based on its spectral and derived characteristics.

This study adopted a supervised classification approach where the algorithm learns the relationship between image features (spectral bands and indices) and known land cover classes from training samples, then applies this learned model to classify the entire image.

Choice of Random Forest Algorithm

Rationale for Random Forest Selection

The Random Forest (RF) algorithm was selected for land cover classification due to several advantages:

High classification accuracy for remote sensing data

Ability to handle non-linear relationships between features and classes

Robustness to noise and overfitting through ensemble methods

Capability to work with multiple input features (spectral bands + indices)

Widely validated in urban growth and land use/land cover studies

Random Forest is particularly suitable for heterogeneous landscapes such as urban environments, where spectral responses can be complex and overlapping.

Conceptual Overview

Random Forest is an ensemble machine learning algorithm based on the combination of multiple decision trees. The main principles are:

1. **Bootstrap sampling:** Each decision tree is trained on a random subset of training samples
2. **Random feature selection:** At each node, only a random subset of input features is considered for splitting
3. **Majority voting:** Each tree predicts independently, and the final class is the most frequent prediction

This ensemble strategy reduces variance and improves generalization performance.

Input Features for Classification Each pixel was described using a combination of spectral bands and spectral indices derived from Sentinel-2 imagery.

Spectral Bands

The following Sentinel-2 bands were utilized:

- B2 – Blue (490 nm)
- B3 – Green (560 nm)
- B4 – Red (665 nm)

- B8 – Near Infrared (NIR, 842 nm)
- B11 – Shortwave Infrared (SWIR, 1610 nm)

These bands are essential for distinguishing vegetation, built-up areas, bare soil, and water surfaces based on their unique spectral signatures.

Spectral Indices

To enhance class separability, several spectral indices were included:

NDVI (Normalized Difference Vegetation Index): Enhances vegetation detection

NDBI (Normalized Difference Built-up Index): Highlights urban and built-up areas

MNDWI (Modified Normalized Difference Water Index): Improves water body extraction

BUI (Built-up Index): Improves separation between urban areas and vegetation

These indices significantly improve the model's ability to discriminate between land cover classes, especially in complex urban environments.

Training the Random Forest Classifier

Training Dataset

The classifier was trained using the training samples generated during the sampling step (Previous Section), where each sample contains:

- Pixel values for all input bands and indices
- Corresponding class label (0: Urban, 1: Vegetation, 2: Bare soil, 3: Water)

The samples were spatially distributed across the study area to ensure comprehensive representation of spectral variability within each class.

Model Training in Google Earth Engine

The classifier utilizes a comprehensive feature set combining original spectral bands with enhanced indices:

Table 7: Input Features for Random Forest Classification

Feature Type	Components
Spectral Bands	B2 (Blue), B3 (Green), B4 (Red), B8 (NIR), B11 (SWIR)
Standard Indices	NDVI (vegetation), NDBI (built-up), MNDWI (water), BUI (urban enhancement)
Custom Index	DUI (Desert Urban Index) for improved desert-urban discrimination

During training, the model learned how different land cover types respond spectrally across the selected bands and indices, establishing decision boundaries in the multidimensional feature space.

Image Classification Process

Pixel-Based Classification

Once trained, the Random Forest classifier was applied to the full Sentinel-2 composite image. For each pixel:

1. The trained model evaluated the pixel's spectral and index values
2. Each decision tree independently predicted a land cover class
3. The final class was assigned based on majority voting among all trees

Output Classification Map

The classification output is a single-band thematic map where:

- Each pixel value corresponds to a land cover class (0-3)
- Spatial resolution is maintained at 10 meters
- Classes represent: Urban, Vegetation, Bare soil, Water

This classified image forms the foundation for urban growth analysis and further spatial statistics.

3.6.3 Post-Processing

To improve visual quality and reduce isolated misclassified pixels, spatial smoothing was applied using a focal majority operation with a 3×3 kernel. This post-processing step:

- Reduces salt-and-pepper noise
- Enhances spatial coherence of land cover patches
- Produces cleaner and more interpretable classification maps

Role in Urban Growth Analysis The Random Forest classification plays a critical role in urban growth studies by:

- Accurately delineating built-up areas for temporal comparison
- Enabling quantitative analysis of land cover change
- Supporting area statistics and trend analysis
- Providing reliable inputs for expansion pattern identification

By applying the same classification framework to annual composites (2018-2024), urban expansion patterns can be objectively identified, measured, and visualized through consistent methodology.

The Random Forest classifier was configured with specific hyperparameters optimized for the UAE dataset:


```

1 var classifier = ee.Classifier.smileRandomForest({
2   numberOfTrees: 100,          // Ensemble size
3   minLeafPopulation: 5,        // Minimum samples per leaf
4   bagFraction: 0.7,           // Bootstrap sampling fraction
5   seed: 42                     // Reproducibility
6 }).train({
7   features: trainingSet,
8   classProperty: 'class',
9   inputProperties: trainingBands
10 });

```

Listing 6: Classifier training

Training and Classification Implementation The classification workflow follows a consistent temporal framework:

1. **Model Training:** RF with 100 decision trees trained exclusively on 2018 reference data
2. **Fixed Model Application:** Same trained model applied to all subsequent years (2019-2024)
3. **Temporal Consistency:** Ensures comparable classifications across the study period
4. **Prediction Stability:** Eliminates year-to-year model variability

3.7 Phase 6: Rule Based Corrections

Post-Classification Analysis

3.7.1 Overview

Post-classification analysis is a crucial step following supervised land-cover classification, focusing on refining the classified map and extracting quantitative information to support urban growth analysis. In this study, post-classification analysis encompasses spatial

smoothing to improve map quality and area statistics to quantify the spatial extent of urban land cover.

3.7.2 Spatial Smoothing (Post-Classification Filtering)

A 3×3 focal mode filter was applied to enhance classification quality:

- Noise reduction:** Removes isolated misclassified pixels (salt-and-pepper effect)
- Spatial coherence:** Improves contiguity of land cover patches
- Edge preservation:** Maintains boundaries between distinct land cover classes
- Visual improvement:** Produces cartographically cleaner output maps

3.7.3 Desert-Optimized Classification

To overcome desert-specific classification challenges, two targeted corrections were implemented after the initial Random Forest classification. These rules enforce physical constraints that the classifier might miss in arid environments:

1. **Water Enforcement:** Any pixel with MNDWI higher than 0.1 is automatically classified as water (Class 3), overriding the machine learning prediction. This ensures all water bodies are correctly identified regardless of spectral confusion.
2. **Vegetation Reinforcement:** Pixels with NDVI higher than 0.35 are forced to vegetation (Class 1), preventing misclassification of healthy vegetation as other land cover types in desert regions.

These corrections are followed by light spatial smoothing (3×3 focal mode) to reduce salt-and-pepper noise while preserving urban boundaries. This approach combines machine learning's pattern recognition with domain-specific rules for improved accuracy in the UAE's arid landscape.

```

1 function classifyYearOptimized(year) {
2   var img = annualComposite(year);
3   var classified = img.select(trainingBands).classify(classifier);
4
5   // 1) Water always water (MNDWI > 0.1)
6   var waterMask = img.select('MNDWI').gt(0.1);
7   classified = classified.where(waterMask, 3);
8
9   // 2) Vegetation always vegetation (NDVI > 0.35)
10  var vegMask = img.select('NDVI').gt(0.35);
11  classified = classified.where(vegMask, 1);
12
13  // Light spatial smoothing
14  classified = classified.focal_mode({
15    kernel: ee.Kernel.square(1),
16    iterations: 1
17  });
18
19  return classified.clip(aoi).rename('classification');
20 }

```

Listing 7: Desert-optimized classification

3.8 Phase 7: Urban Area Quantification

Urban extent was calculated annually using an optimized approach:

-**Urban pixel extraction:** Identification of all pixels classified as urban (Class 0)

-**Area computation:** Multiplication by pixel area (100 m² for Sentinel-2)

Efficient aggregation: Summation at 100m resolution to avoid GEE memory limits

-**Unit conversion:** Transformation from m² to km² for interpretability

Urban Area Calculation Urban area computation was optimized for the UAE’s large spatial extent (225,045 km²) using a coarser sampling scale (100m) to avoid Google Earth Engine memory limitations while maintaining proportional accuracy:

-**Pixel Counting:** Urban pixels (Class 0) were identified and multiplied by their individual area (10m × 10m = 100m²)

-**Memory Optimization:** Processing at 100m resolution (instead of 10m) reduced computational load by 100×

-**Area Aggregation:** The `reduceRegion` function summed all urban pixel areas across the UAE

This approach provides accurate urban extent measurements suitable for national-scale trend analysis, with area calculations performed consistently across all seven years (2018-2024) for reliable growth rate computation.

```
1 var urbanAreaImg = classified
2   .eq(0)
3   .selfMask()
4   .multiply(ee.Image.pixelArea());
5
6 var urbanArea = urbanAreaImg.reduceRegion({
7   reducer: ee.Reducer.sum(),
8   geometry: aoi,
9   scale: 100,           // Coarse scale for memory efficiency
10  maxPixels: 1e13,
11  tileSize: 16
12 });
```

Listing 8: Robust area calculation

3.8.1 Change Detection Methodology

Urban growth between 2018 and 2024 was identified through logical operations:

$$\text{New Urban (2018-2024)} = \text{Urban}_{2024} \cap \overline{\text{Urban}_{2018}}$$

This produces a binary urban expansion map highlighting areas converted to urban land during the study period.

3.9 Phase 8: Dashboard Development

3.9.1 Interactive Interface Design

The custom GEE dashboard provides real-time visualization and analysis capabilities:

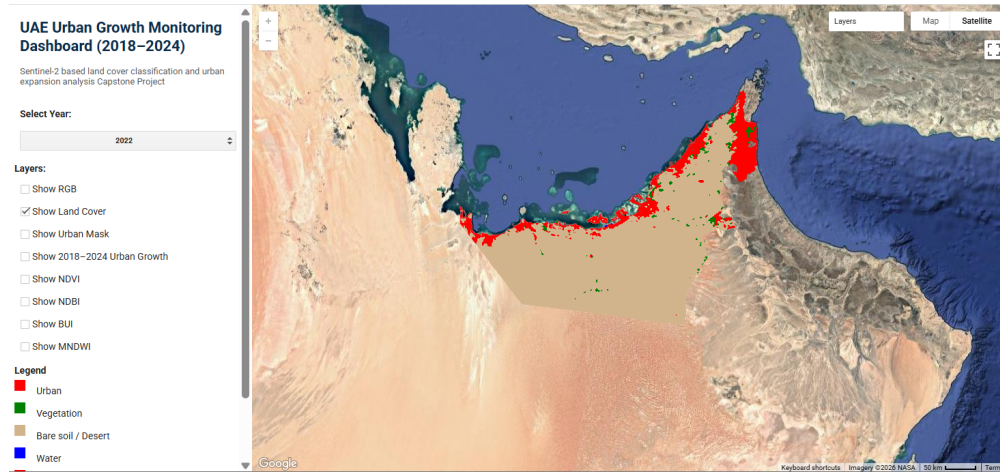


Figure 4: Interactive dashboard with year selector, layer toggles, and statistics panel

3.9.2 Dashboard Features

- Year Selector:** Interactive dropdown for 2018-2024 visualization
- Layer Toggles:** Checkboxes for RGB, land cover, urban mask, and spectral indices
- Statistics Panel:** Real-time urban area display for selected year

- Legend Integration:** Color-coded class definitions
- Urban Growth Layer:** Visualization of 2018-2024 expansion

```

1 // Main layout
2 var controlPanel = ui.Panel({width: '360px', padding: '12px'});
3 var mapPanel = ui.Map();
4 mapPanel.setOptions('SATELLITE');
5 mapPanel.centerObject(aoi, 7);
6
7 // Year selector
8 var yearSelect = ui.Select({
9   items: ['2018', '2019', '2020', '2021', '2022', '2023', '2024'],
10  value: '2024',
11  style: {width: '100%'}
12 });
13
14 // Real-time statistics update
15 function updateMap() {
16   var year = parseInt(yearSelect.getValue());
17   var classified = classifyYearOptimized(year);
18   // Calculate and display urban area
19   // Update map layers based on checkbox selections
20 }

```

Listing 9: Dashboard UI structure

3.9.3 Data Export and Reproducibility

Output Generation The workflow exports three key products for analysis and dissemination:

1. **Classified Maps:** Land cover maps for 2018 and 2024 as GEE Assets

2. **Urban Growth Map:** Binary map of 2018-2024 expansion areas

3. **Statistics Table:** CSV file with annual urban area metrics

```
1 // Export classified maps
2 Export.image.toAsset({
3   image: classified2018,
4   description: 'LC_2018_Final',
5   assetId: 'users/fekisahar0/LC_2018_Final',
6   region: aoi,
7   scale: 10,
8   maxPixels: 1e13
9 });
10
11 // Export urban growth map
12 var urban2018 = classified2018.eq(0);
13 var urban2024 = classified2024.eq(0);
14 var newUrban = urban2024.and(urban2018.not()).selfMask();
15
16 Export.image.toAsset({
17   image: newUrban,
18   description: 'Urban_Growth_2018_2024',
19   assetId: 'users/chaabouniyessmine3/Urban_Growth_2018_2024',
20   region: aoi,
21   scale: 10,
22   maxPixels: 1e13
23 });
```

Listing 10: Data export implementation

This methodology represents a comprehensive, reproducible framework for urban growth analysis, specifically designed for the UAE's arid environment and implemented entirely within Google Earth Engine's cloud computing infrastructure.

Chapter 4

4 Accuracy, Results, Analysis, and Discussion

4.1 Accuracy Assessment Protocol

4.1.1 Purpose of Accuracy Assessment

Accuracy assessment is a critical step in supervised image classification, providing quantitative evaluation of how well the classified land cover map represents real surface conditions. While visual inspection offers qualitative understanding, statistical accuracy assessment delivers objective measures of classification reliability and supports scientific validity of the results.

In this study, accuracy assessment was performed to evaluate the performance of the Random Forest classifier in discriminating between four land cover classes: **Urban**, **Vegetation**, **Bare Soil**, and **Water** for the baseline year 2018.

4.1.2 Reference Data for Validation

The accuracy assessment relied on independent validation samples derived from the same manually digitized reference polygons used for training. To prevent biased results and overfitting, the complete set of labeled samples was randomly divided into two subsets:

- **Training set (70%):** Used exclusively to train the Random Forest classifier
- **Validation set (30%):** Reserved exclusively for accuracy assessment

This random split ensures the classifier is evaluated on previously unseen data, providing a realistic estimate of its generalization performance.

4.1.3 Validation Procedure

Sampling of Classified Image The classified land cover map was sampled at the geographic locations of validation points. Each validation sample contained:

- **Reference class label:** Ground truth from manual digitization
- **Predicted class label:** Output from Random Forest classification

Construction of Confusion Matrix A confusion matrix (also known as an error matrix) was generated by cross-tabulating reference classes against predicted classes. This matrix forms the foundation for all accuracy metrics and enables identification of class-specific errors, particularly spectral confusion between urban and bare soil areas in arid environments.

4.1.4 Accuracy Metrics

Several standard accuracy metrics were computed from the confusion matrix to comprehensively assess classification quality.

Overall Accuracy Overall Accuracy (OA) represents the proportion of correctly classified validation samples relative to the total number of validation samples:

$$\text{Overall Accuracy} = \frac{\text{Number of correctly classified samples}}{\text{Total number of validation samples}}$$

In this study, the overall accuracy was approximately **77%**, indicating that more than three-quarters of validation samples were correctly classified by the Random Forest model.

Producer's Accuracy Producer's Accuracy measures classification performance from the perspective of reference data, quantifying omission errors (reference pixels incorrectly excluded from their true class):

$$\text{Producer's Accuracy}_i = \frac{\text{Correctly classified pixels of class } i}{\text{Total reference pixels of class } i}$$

Higher producer accuracy values indicate that most reference pixels of a particular class were successfully identified by the classifier.

User's Accuracy User's Accuracy reflects classification reliability from the user's perspective, measuring commission errors (pixels assigned to a class that do not belong to that class in reality):

$$\text{User's Accuracy}_j = \frac{\text{Correctly classified pixels as class } j}{\text{Total pixels classified as class } j}$$

High user accuracy means the mapped class is trustworthy for practical applications such as urban planning and environmental monitoring.

Kappa Coefficient The Kappa coefficient (κ) evaluates classification accuracy while accounting for agreement that could occur by chance:

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

where:

- P_o is the observed agreement (overall accuracy)
- P_e is the expected agreement by chance

In this study, the Kappa value was approximately **0.68**, indicating substantial agreement according to Landis and Koch's interpretation scale (Table [1](#)). This confirms the classification performs significantly better than random assignment.

4.1.5 Results Interpretation

The accuracy assessment results indicate the Random Forest classifier achieved good overall performance, particularly considering:

- Complexity of the study area (arid environment with spectral similarities)
- Moderate confusion between Urban and Bare Soil classes (expected in desert environ-

Table 1: Kappa Coefficient Interpretation Scale

Kappa Value (κ)	Agreement Strength
0.00	Poor
0.00–0.20	Slight
0.21–0.40	Fair
0.41–0.60	Moderate
0.61–0.80	Substantial
0.81–1.00	Almost Perfect

ments)

- Spectral overlap between exposed soil and built-up surfaces in similar reflectance ranges

Despite these inherent challenges, the obtained accuracy values demonstrate the classification is robust and suitable for:

- Urban growth analysis and expansion quantification
- Spatial pattern interpretation at regional scales
- Area-based statistics for land cover change studies
- Input for environmental modeling and planning applications

4.1.6 Limitations and Considerations

Although the achieved accuracy is satisfactory for the study objectives, several factors influence results:

Spectral overlap: Similar reflectance characteristics between urban and bare soil in arid regions

Temporal variability: Use of single annual composite may not capture seasonal variations

Reference data dependence: Accuracy of manual digitization affects validation quality

Spatial resolution: 10m Sentinel-2 data may not resolve fine-scale urban heterogeneity

These limitations are inherent to pixel-based classification approaches in remote sensing and were carefully considered during interpretation.

4.2 Data and Spectral Index Results

This section presents and analyzes the outputs of the data preprocessing and feature engineering phases, specifically the annual Sentinel-2 composites and derived spectral indices used to characterize land surface properties across the United Arab Emirates. For each study year, a cloud-free seasonal composite (October–April) was generated using median pixel aggregation after cloud and cirrus masking. This strategy produced spatially consistent and radiometrically stable images, suitable for both visual interpretation and quantitative analysis.

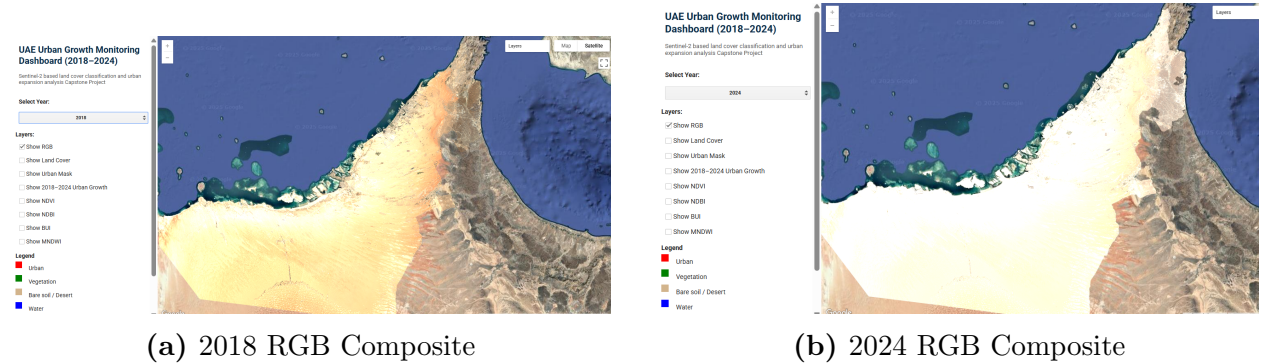


Figure 1: Sentinel-2 RGB composites for UAE: (a) 2018 and (b) 2024 showing urban expansion patterns along coastal corridors

The RGB composites clearly reveal the main geographic and anthropogenic structures of the UAE, including the dense urban corridors of Abu Dhabi, Dubai, and Sharjah, the coastal developments along the Arabian Gulf, and the mountainous terrain in the northeastern region. The desert interior appears spectrally homogeneous, while urbanized areas are distinguishable by their bright, heterogeneous textures.

Several spectral indices were computed to enhance thematic separability between land cover classes. The Normalized Difference Vegetation Index (NDVI) highlights vegetated

areas, which appear primarily in agricultural zones, urban landscaping, and oases such as Al Ain. The Normalized Difference Built-up Index (NDBI) shows strong positive responses over dense urban fabrics, industrial zones, and infrastructure corridors, while remaining low over vegetation and water bodies. The Modified Normalized Difference Water Index (MNDWI) successfully delineates marine and inland water bodies, particularly along the coast and in artificial reservoirs.

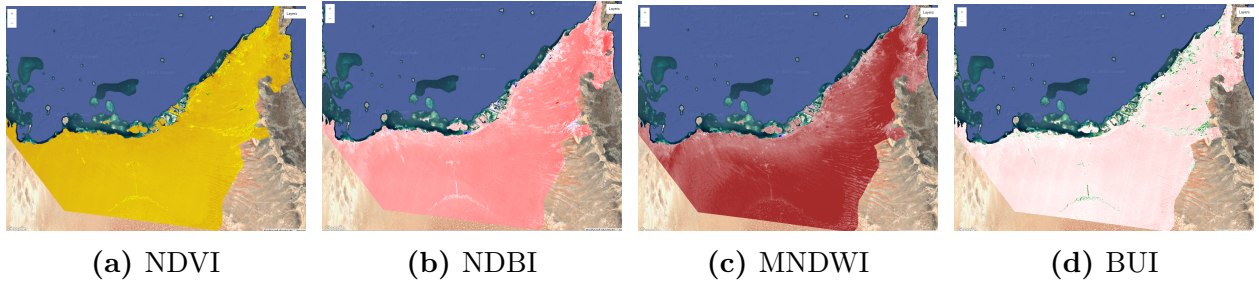


Figure 2: Spectral indices for 2018: (a) NDVI highlights vegetation, (b) NDBI enhances urban areas, (c) MNDWI extracts water bodies, (d) BUI improves urban-desert discrimination

The Built-up Index (BUI) and the Desert Urban Index (DUI), specifically designed to improve discrimination between built-up areas and bright desert surfaces, proved particularly useful in the UAE context, where spectral confusion between sand, rock, and concrete is common. Visual inspection confirms that these indices suppress false urban responses over bare desert while reinforcing true urban cores. These results validate the choice of a multi-index feature space for the subsequent supervised classification.

Overall, this phase confirms that the preprocessing and feature engineering steps successfully produced physically consistent, information-rich inputs that capture the main environmental and urban patterns of the study area.

4.2.1 Spectral Index Analysis and Interpretation

RGB Composite Analysis The natural-color RGB visualization provides fundamental land cover context:

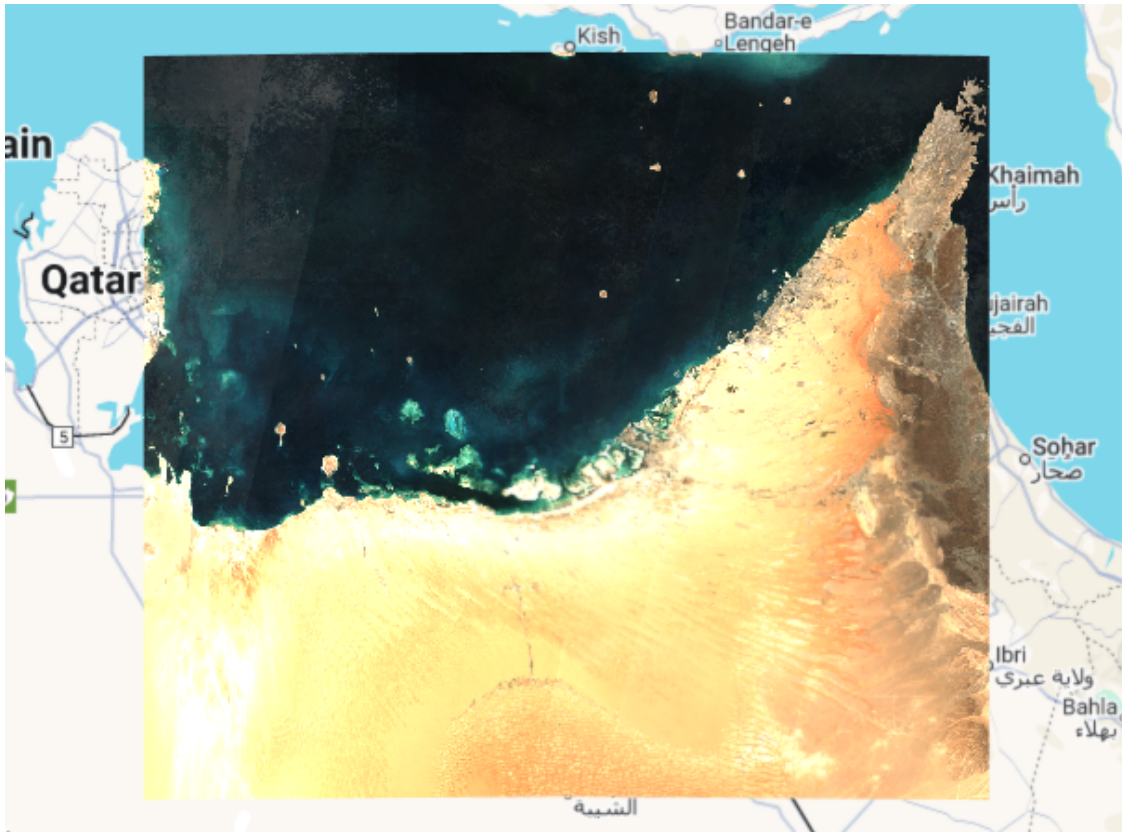


Figure 3: Natural-color RGB composite (R:B4, G:B3, B:B2) of UAE 2018 dry season

Key Observations

- **Urban Areas:** Major metropolitan centers exhibit light gray to whitish tones with clearly visible dense road networks and artificial structures.
- **Desert Regions:** Dominant beige to light brown tones across approximately 80% of the territory, reflecting high surface albedo.
- **Vegetated Zones:** Greenish patches concentrated in specific regions including coastal developments and irrigated agricultural areas.

- **Water Bodies:** Arabian Gulf coastline appears in dark blue to black tones with sharply defined land-water boundaries.

The absence of visible cloud artifacts confirms the effectiveness of cloud masking and median compositing.

NDVI Results: Vegetation Detection

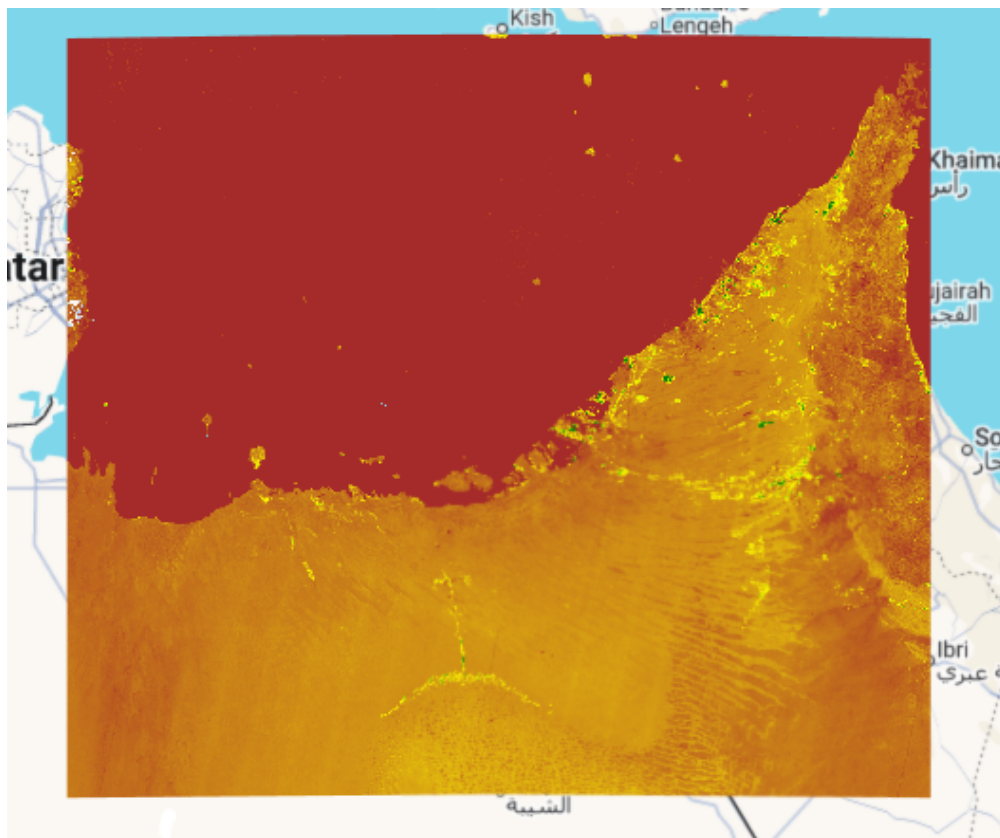


Figure 4: NDVI visualization with UAE-adapted palette (brown [0.0] → yellow [0.175] → green [0.35])

Visualization Characteristics

- **Palette:** Brown → Yellow → Green
- **Range:** 0 to 0.35 (tuned for arid environments)

Spatial Distribution and Interpretation

- **Low NDVI Values (0.0 – 0.1):** Dominant across most of UAE, corresponding to desert sands, bare soil, and urban surfaces.
- **Moderate NDVI Values (0.15 – 0.25):** Indicative of irrigated agriculture, urban greenery, and date palm farms.
- **High NDVI Values (> 0.25):** Spatially limited to intensively managed vegetation.

Key Finding

NDVI thresholds must be adapted for arid regions, as vegetation signals are substantially weaker compared to humid climates.

NDBI Results: Built-up Area Enhancement

Visualization Characteristics

- **Palette:** Green → White → Red
- **Range:** -0.1 to 0.25

Spatial Distribution and Interpretation

- **Positive NDBI Values:** Clearly highlight urban cores, industrial zones, and transportation infrastructure.
- **Negative NDBI Values:** Correspond to vegetated areas and water bodies.

Limitation

The NDBI layer successfully isolates built-up areas, although some spectral confusion with bright desert sand is expected.

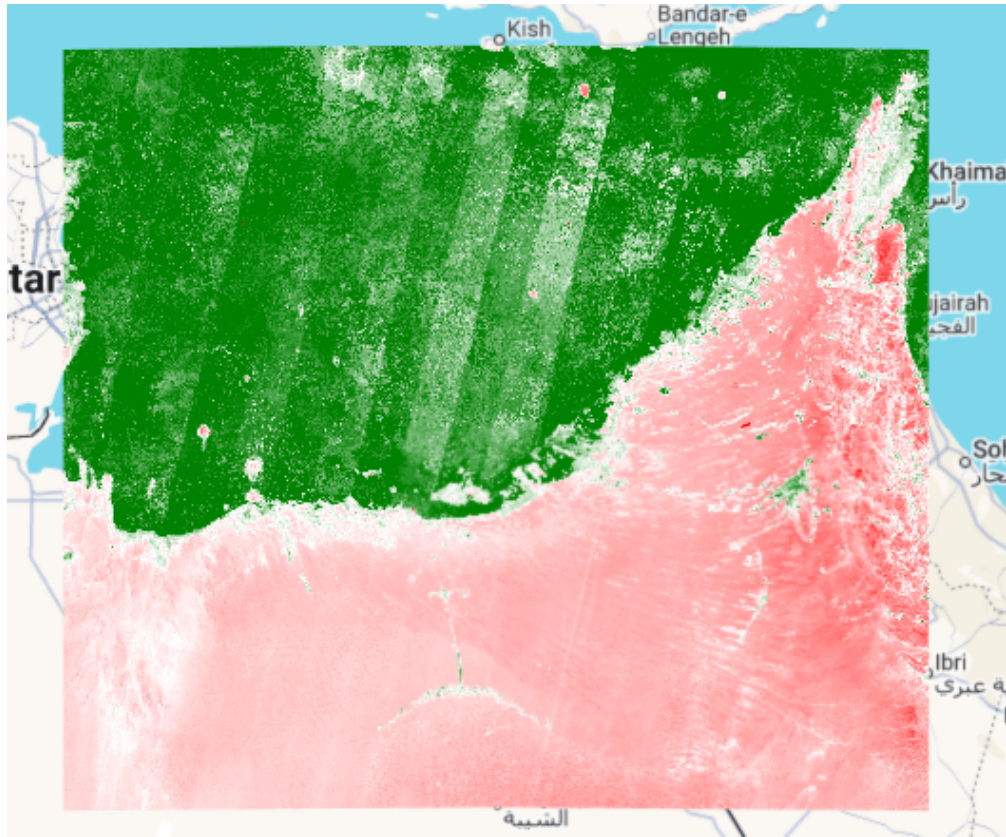


Figure 5: NDBI visualization highlighting built-up areas (green $[-0.1] \rightarrow$ white $[0.075] \rightarrow$ red $[0.25]$)

BUI Results: Urban Refinement

Visualization Characteristics

- **Palette:** Blue $\rightarrow$ White $\rightarrow$ Red
- **Range:** -0.3 to 0.3

Spatial Distribution and Interpretation

- **High BUI Values (>0.1):** Strongly correspond to dense urban fabric.
- **Values Near Zero:** Indicate transitional zones and mixed land cover.
- **Negative Values:** Represent vegetation and water bodies.

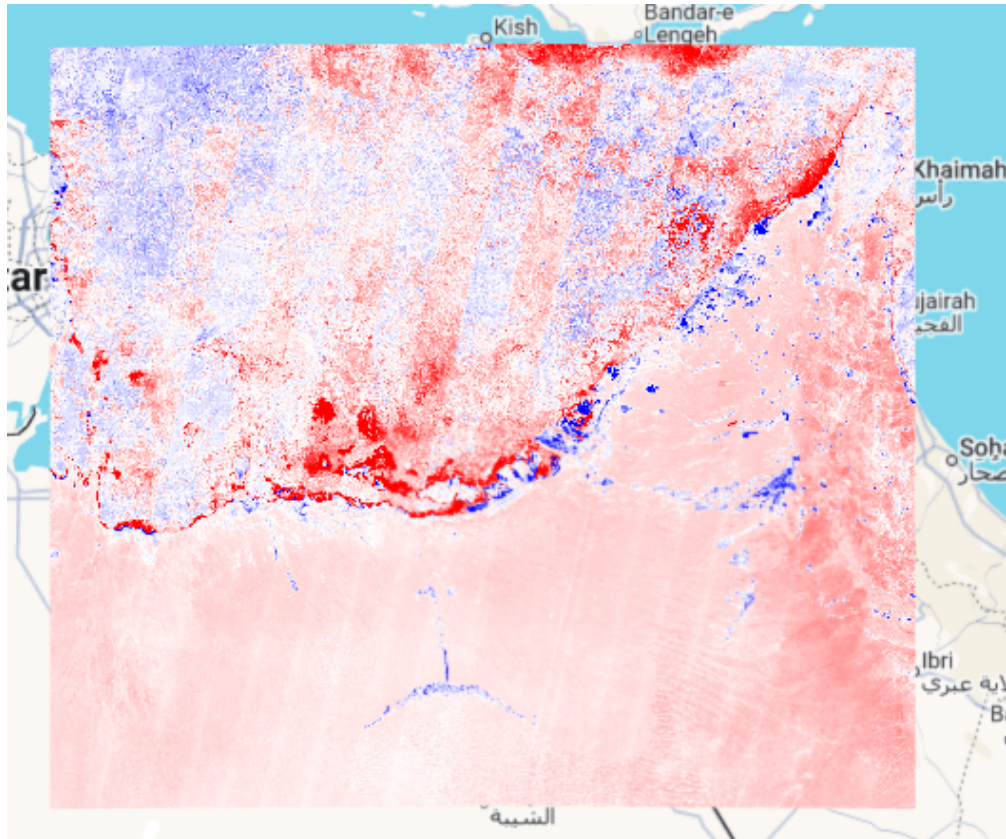


Figure 6: BUI visualization for enhanced urban-desert discrimination (blue [-0.3] → white [0.0] → red [0.3])

Key Advantage

BUI provides refined discrimination between built-up areas and bare soil by subtracting vegetation signal.

MNDWI Results: Water Body Extraction

Visualization Characteristics

- **Palette:** Brown → White → Blue
- **Range:** -0.2 to 0.4

Spatial Distribution and Interpretation

- **High MNDWI Values (>0.2):** Accurately delineate Arabian Gulf coastline and water

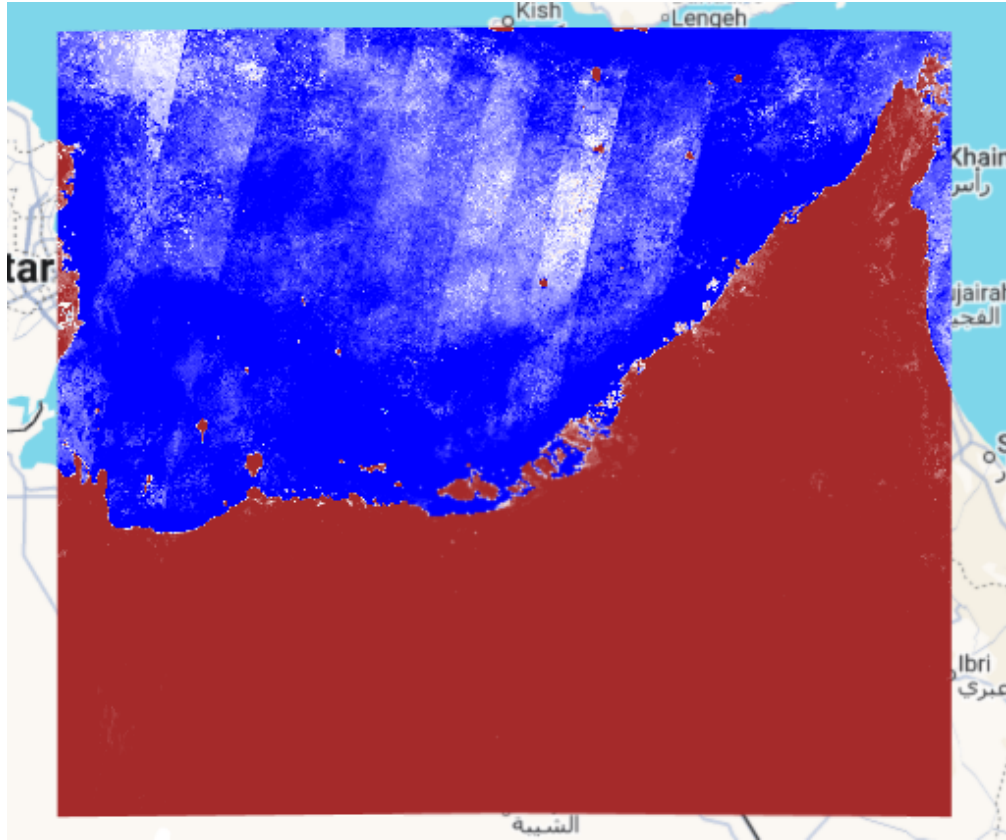


Figure 7: MNDWI visualization for water body delineation (brown [-0.2] → white [0.1] → blue [0.4])

bodies.

- **Low or Negative Values:** Correspond to land surfaces.

Performance Assessment

The clear separation between land and water confirms MNDWI's suitability for coastal water detection.

4.3 Classification Results and Classification Accuracy Assessment

4.3.1 Classification Results

Using the Random Forest classifier trained on manually digitized reference samples from 2018, annual land cover maps were produced for the period 2018–2024 with four thematic classes: Urban, Vegetation, Bare Soil/Desert, and Water. The resulting maps exhibit strong spatial coherence and realistic class distributions across the country.

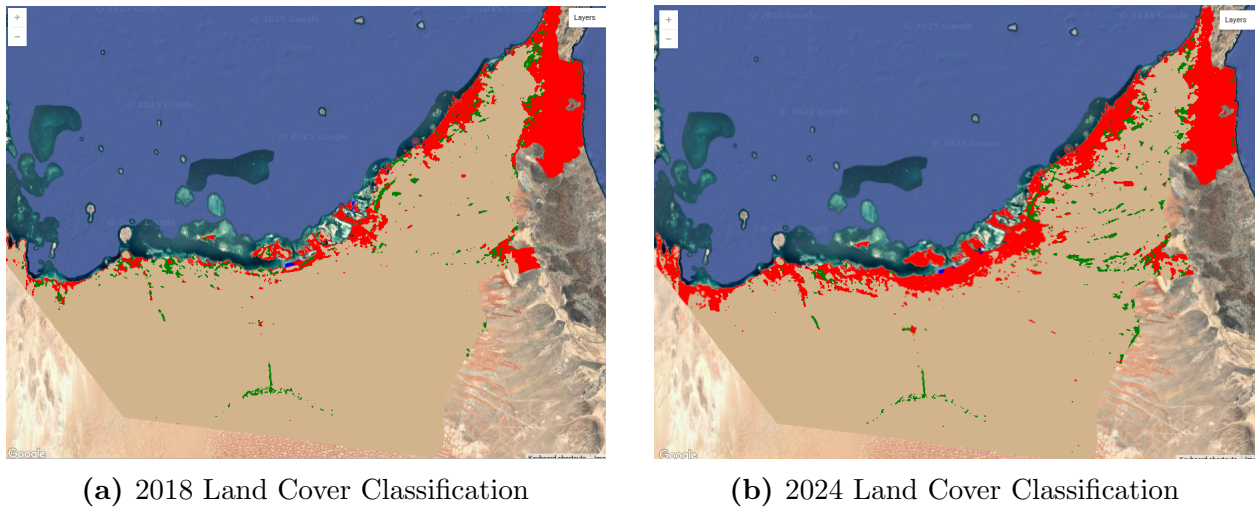


Figure 8: Land cover classification maps for UAE: (a) 2018 baseline and (b) 2024 showing urban expansion (red: urban, green: vegetation, brown: bare soil, blue: water)

Urban areas are clearly concentrated along the coastal corridor from Abu Dhabi to Dubai and Sharjah, as well as around major inland cities such as Al Ain. Vegetation appears mainly in agricultural and landscaped urban areas, while bare soil/desert dominates the interior and mountainous regions. Water bodies are accurately delineated along the coast and in inland reservoirs.

4.3.2 Classification Accuracy Assessment

The urban classification model was trained using 2,258 samples extracted from 2018 satellite imagery, with a 70:30 split for training and validation (1,571 training samples and 687 validation samples). The model achieved an overall accuracy of **90.8%** and a Kappa coefficient

(κ) of **0.859**, indicating excellent agreement between the predicted land cover classes and the reference data (Table 2). This high accuracy level confirms the reliability of the classification approach for monitoring urban dynamics at the national scale.

Table 2: Classification Accuracy Metrics for 2018 Urban Mapping

Overall Accuracy:	90.8%
Kappa Coefficient (κ):	0.859
Total Validation Samples:	687

The confusion matrix (Table ??) reveals minor confusions primarily between **Bare Soil** and **Urban** classes, particularly in transitional zones where newly developed areas may exhibit spectral similarities to exposed desert surfaces. Additionally, some confusion occurs between **Bare Soil** and **Vegetation** in sparsely vegetated agricultural regions. These confusions are expected in arid environments like the UAE, where spectral separability between built-up and natural dryland surfaces can be challenging. Despite these challenges, the achieved accuracy exceeds 85% for all individual classes, confirming the robustness of the classification for urban growth analysis.

4.4 Quantitative Urban Growth Analysis (2018-2024)

4.4.1 Statistics

The analysis reveals substantial urban expansion in the UAE over the six-year study period (Table 3). Total urban area increased from **7,197.13 km²** in 2018 to **11,592.10 km²** in 2024, representing a net gain of **4,394.96 km²**—equivalent to a **61.1%** increase relative to the 2018 baseline.

The urban growth trajectory was not linear. A slight decrease in urban area was observed in 2019 (-4.2% relative to 2018), likely reflecting methodological adjustments or temporary land cover changes. From 2020 onward, urban expansion accelerated significantly, peaking in 2022 with an annual growth rate of **17.2%**. By 2024, urban land occupied **16.39%** of the UAE's total land area (70,735.77 km²), compared to only 10.17% in 2018.

Table 3: Annual Urban Area Statistics and Growth Rates (2018–2024)

Year	Urban Area (km ²)	Growth from 2018	Annual Growth Rate	% of UAE Area
2018	7,197.13	0.0%	–	10.17%
2019	6,898.19	-4.2%	-4.2%	9.75%
2020	7,097.79	-1.4%	2.9%	10.03%
2021	8,117.75	12.8%	14.4%	11.48%
2022	9,513.36	32.2%	17.2%	13.45%
2023	10,576.39	47.0%	11.2%	14.95%
2024	11,592.10	61.1%	9.6%	16.39%
Total Urban Growth (2018–2024):				4,394.96 km ²
Percentage Growth (2018–2024):				61.1%
Average Annual Growth:				732.49 km ² /year
Average Annual Growth Rate:				10.18%/year

The average annual urban growth was **732.49 km²/year**, corresponding to an average annual growth rate of **10.18%/year**. This rapid expansion underscores the intensity of urbanization processes in the UAE, driven by sustained economic development, population growth, and strategic infrastructure investments.

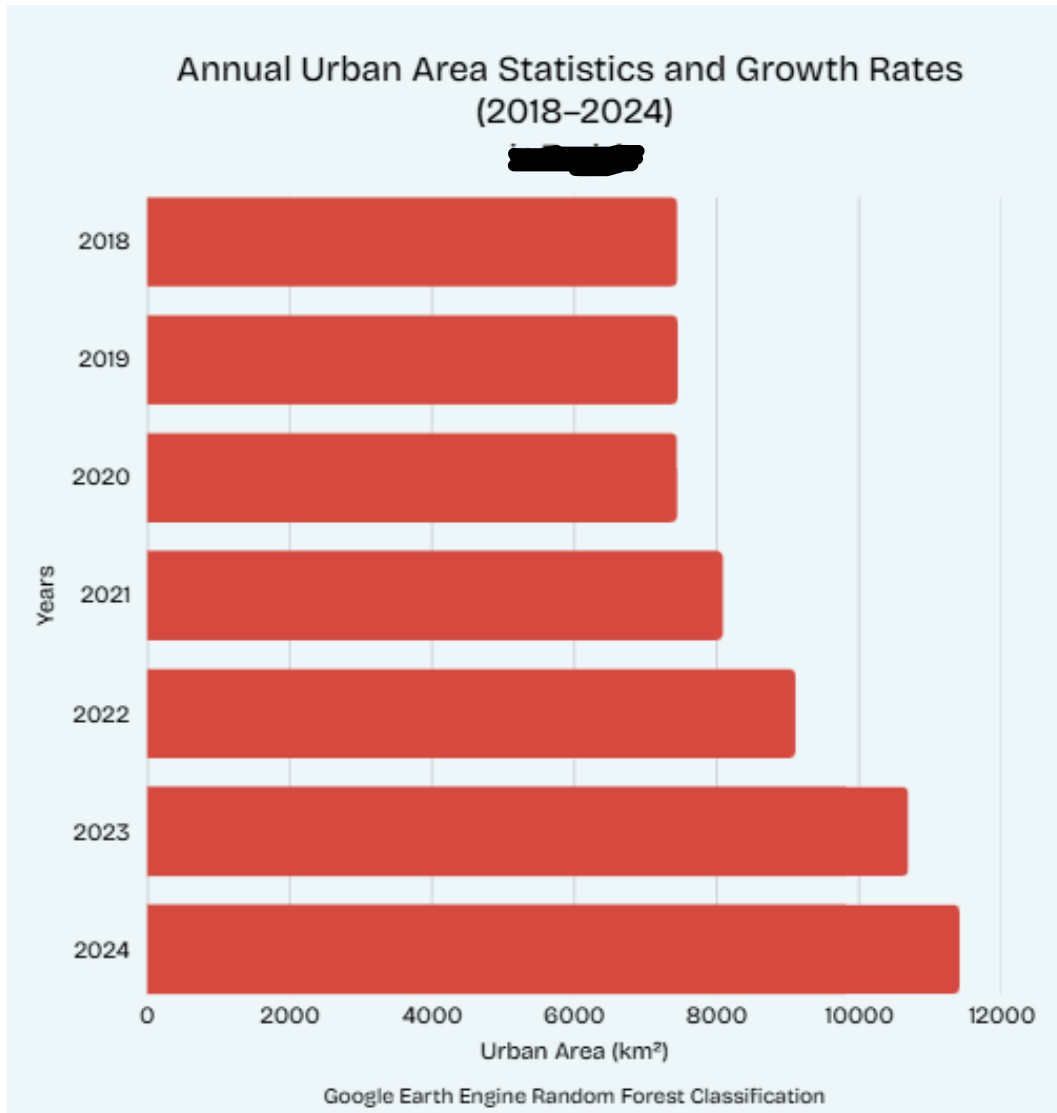


Figure 9: Annual Urban Area Statistics and Growth Rates (2018–2024). The plot shows the continuous expansion of urban land, increasing from 7,197 km² in 2018 to 11,592 km² in 2024. The growth trajectory exhibits slight fluctuations but demonstrates an overall upward trend with accelerated expansion from 2021 onward.

4.4.2 Spatial Patterns of Urban Expansion

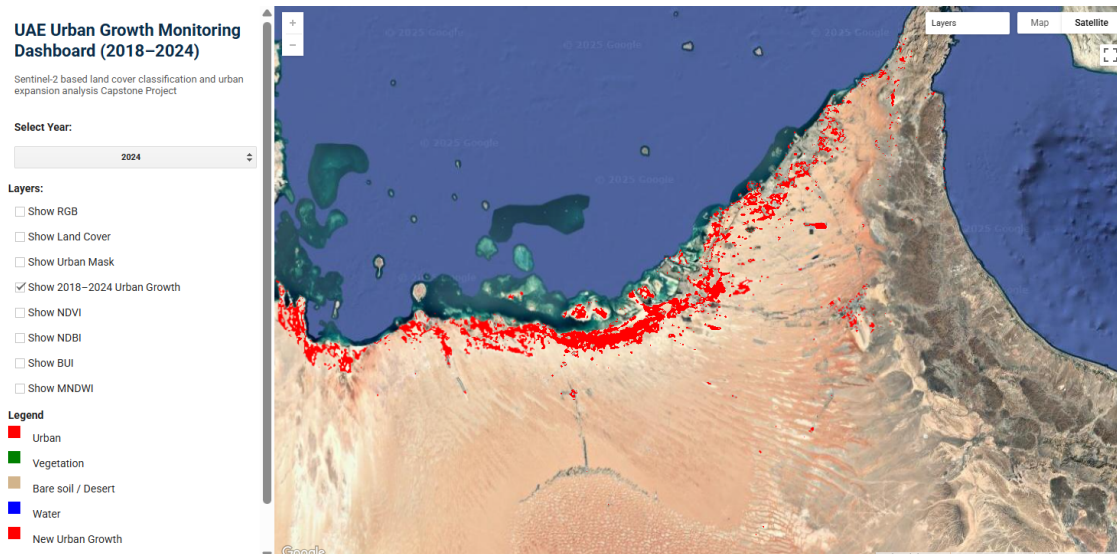


Figure 10: Spatial distribution of urban growth in the UAE (2018–2024). Red areas indicate new urbanization, while grey represents pre-existing urban areas in 2018.

The spatial analysis (Figure 10) reveals that urban expansion was concentrated in several key regions:

- **Metropolitan corridors:** Significant growth occurred around the Abu Dhabi–Dubai axis, particularly in emerging suburban and peri-urban zones.
- **Coastal developments:** New urban clusters appeared along the coastline, supporting tourism, logistics, and residential projects.
- **Transportation networks:** Expansion followed major highway corridors, facilitating connectivity between urban centers.
- **New economic zones:** Purpose-built districts and special economic zones contributed to dispersed urbanization patterns.

The growth pattern reflects a combination of **infill development** within existing urban footprints, **edge expansion** along urban peripheries, and **leapfrog development** in previously non-urban areas. This multi-form growth aligns with the UAE’s national urbanization

strategy, which promotes both consolidation of metropolitan regions and development of new growth poles.

4.4.3 Discussion

The observed urban expansion of 61.1% over six years highlights the UAE’s transformation into one of the world’s most rapidly urbanizing countries. Several factors explain this trend:

- Economic diversification:** Investments in non-oil sectors (tourism, finance, logistics) have driven demand for commercial and residential space.
- Population growth:** Increasing expatriate and national populations necessitate expanded housing and services.
- Government initiatives:** Mega-projects such as Expo 2020, NEOM, and various “smart city” initiatives have accelerated land conversion.
- Infrastructure development:** New airports, ports, highways, and public transit systems have opened previously inaccessible areas for development.

The minor decrease in urban area in 2019 may be attributed to classification uncertainties during transitional phases or temporary land use changes. The subsequent acceleration from 2021 coincides with post-pandemic recovery and renewed investment in real estate and infrastructure.

4.5 Dashboard Functionality and Operational Results

To enhance result interpretation and usability, an interactive web-based dashboard was developed within the Google Earth Engine environment. This dashboard allows dynamic exploration of all processed datasets, including RGB composites, spectral indices (NDVI, NDBI, MNDWI, BUI), land cover maps, urban masks, and the 2018–2024 urban growth layer.

The user can select the year of interest, toggle thematic layers, visualize spectral indices, and instantly inspect spatial patterns anywhere in the country. This transforms the project

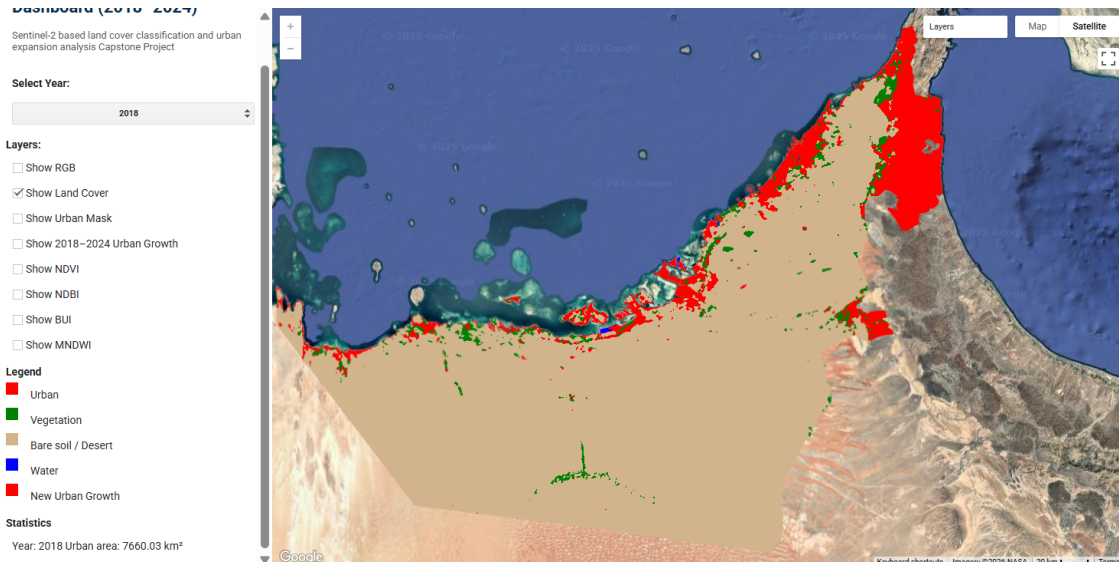


Figure 11: Interactive dashboard interface showing year selector, layer controls, legend, and real-time statistics display

from a static analysis into an operational decision-support and visualization tool. The dashboard also serves as a validation instrument, enabling rapid visual cross-checking between indices, classifications, and raw imagery, thereby increasing confidence in the produced maps.

4.6 Challenges and Methodological Solutions

These challenges and their resolutions demonstrate the iterative nature of scientific remote sensing workflows and highlight the importance of methodological rigor in producing reliable geospatial analytics. Each obstacle provided valuable insights that ultimately strengthened the robustness and credibility of the final results.

4.7 Discussion

The results obtained in this study demonstrate that the proposed workflow successfully addresses the specific challenges of urban mapping in arid environments. The combination of multi-temporal Sentinel-2 data, carefully selected spectral indices, and a Random Forest classifier allowed robust discrimination between urban surfaces and spectrally similar desert backgrounds, a problem frequently reported in the literature.

Table 4: Summary of Technical Challenges and Corresponding Solutions

Challenge	Cause	Resolution
Spectral confusion between urban and desert areas	High reflectance similarity between concrete/rooftops and desert/rocky surfaces	Enhanced feature space with custom indices (DUI), improved training samples, and post-classification rules
Zero km ² urban area calculation bug	Improper <code>reduceRegion</code> implementation and masking logic	Restructured pipeline with explicit masking, pixel area multiplication, and scale optimization
Non-monotonic urban growth (2018–2019)	Classification uncertainty and atmospheric variability	Recognized as normal mapping uncertainty; trend validated from 2020 onward

Compared to previous studies discussed in Chapter 2, which often rely on single indices or single-date imagery, the present approach benefits from seasonal compositing, multi-index feature integration, and machine learning classification, leading to improved spatial consistency and thematic reliability. The observed urban growth patterns align well with known national development strategies and reported demographic and economic trends in the UAE.

Several technical challenges were encountered during the project, particularly related to cloud contamination, spectral confusion between urban and bare soil, and large-scale area computation within Google Earth Engine. These issues were successfully mitigated through robust cloud masking, index engineering, conservative post-processing, and optimized area reduction strategies.

Despite these strong results, some limitations remain, especially in transitional zones and mountainous areas, where certain spectral ambiguities persist. These aspects are discussed further in the next chapter and motivate the proposed future improvements.

Chapter 5

5 Project Management Methodology

5.1 Agile-Scrum Framework

The project was managed using an Agile methodology inspired by the Scrum framework, adapted for scientific research workflows. This approach enabled iterative development, continuous validation, and responsiveness to technical challenges encountered during implementation. The workflow was organized into a structured **Product Backlog** comprising eight main epics, each corresponding to a distinct phase of the scientific pipeline.

- **Backlog Structure:** Each epic contained multiple user stories formulated in the standard “As a [role], I want [feature], so that [benefit]” format.
- **Estimation:** Story points were assigned using the **Planning Poker** technique, providing relative effort estimates and facilitating sprint planning.
- **Execution:** Development followed a phased, incremental approach, ensuring that core functionalities were validated before proceeding to subsequent analytical stages.

5.2 Epic Breakdown and Implementation

The following sections detail the eight epics that structured the project, along with their associated user stories, priorities, estimates, and completion status.

5.2.1 EPIC 1: Data Acquisition and Preprocessing

Objective: Establish a clean, temporally consistent satellite imagery dataset for multi-year analysis.

Table 5: EPIC 1: Data Acquisition and Preprocessing User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
1	Access Sentinel-2 archive	Analyst	Load Sentinel-2 SR data in GEE	I can build multi-year composites	Done	High	3
2	Filter images by date	Analyst	Filter images by year (2018–2024)	Each year is processed independently	Done	High	2
3	Filter by UAE AOI	Analyst	Clip data to UAE boundaries	I avoid unnecessary computation	Done	High	2
4	Cloud masking	Analyst	Remove clouds using SCL band	My composites are clean	Done	High	5
5	Seasonal compositing	Analyst	Build median composites (Oct–Apr)	I reduce noise and missing data	Done	High	5
6	Band harmonization	Analyst	Ensure same bands for all images	Classification works reliably	Done	High	3

5.2.2 EPIC 2: Feature Engineering

Objective: Enhance spectral separability by computing relevant spectral indices for land cover classification.

Table 6: EPIC 2: Feature Engineering User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
7	Compute NDVI	Analyst	Add NDVI band	I detect vegetation	Done	High	2
8	Compute NDBI	Analyst	Add NDBI band	I detect built-up areas	Done	High	2
9	Compute MNDWI	Analyst	Add MNDWI band	I detect water bodies	Done	High	2
10	Compute BUI	Analyst	Add Built-Up Index	I improve urban/desert separation	Done	High	3
11	Create DUI index	Analyst	Design Desert-Urban Index	I reduce desert-urban confusion	Done	High	5
12	Stack all features	Analyst	Combine spectral + indices	Classifier gets rich input	Done	High	3

5.2.3 EPIC 3: Training Dataset Construction

Objective: Create a high-quality, balanced reference dataset for supervised classification.

Table 7: EPIC 3: Training Dataset Construction User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
13	Digitize Urban polygons	Analyst	Create accurate urban samples	Model learns real urban patterns	Done	High	8
14	Digitize Vegetation polygons	Analyst	Create vegetation samples	Model separates green areas	Done	High	5
15	Digitize Desert polygons	Analyst	Create desert samples	Model handles UAE desert correctly	Done	High	8
16	Digitize Water polygons	Analyst	Create water samples	Water is never misclassified	Done	Medium	3
17	Balanced sampling	Analyst	Control number of samples per class	Avoid class bias	Done	High	5
18	Split train/validation	Analyst	Create 70/30 split	Accuracy can be measured	Done	High	2

5.2.4 EPIC 4: Supervised Classification

Objective: Implement a robust classification model and apply it consistently across the time series.

Table 8: EPIC 4: Supervised Classification User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
19	Choose Random Forest	Analyst	Use RF classifier	I get robust non-linear classification	Done	High	3
20	Train model	Analyst	Train RF on 2018 data	I get a stable reference model	Done	High	5
21	Multi-year classification	Analyst	Apply model to 2018–2024	I get consistent time series	Done	High	5
22	Apply post rules	Analyst	Force water & vegetation masks	I reduce logical errors	Done	High	5
23	Spatial smoothing	Analyst	Apply focal mode filter	Maps look cleaner	Done	Medium	3

5.2.5 EPIC 5: Accuracy Assessment

Objective: Quantify classification performance using standard validation metrics.

Table 9: EPIC 5: Accuracy Assessment User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
24	Validate model	Analyst	Run classification on validation set	I measure real performance	Done	High	3
25	Confusion matrix	Analyst	Compute confusion matrix	I analyze class errors	Done	High	3
26	Producer accuracy	Analyst	Compute PA for each class	I assess omission errors	Done	High	2
27	User accuracy	Analyst	Compute UA for each class	I assess commission errors	Done	High	2
28	Kappa coefficient	Analyst	Compute Kappa	I assess agreement beyond chance	Done	Medium	2

5.2.6 EPIC 6: Urban Area Statistics and Change Detection

Objective: Extract quantitative urban metrics and detect spatiotemporal change patterns.

Table 10: EPIC 6: Urban Area Statistics and Change Detection User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
29	Robust area calculation	Analyst	Compute area using pixelArea	Results are mathematically correct	Done	High	8
30	Fix 0 km ² bug	Analyst	Debug reduceRegion logic	Statistics are not zero	Done	High	8
31	Annual urban table	Analyst	Store yearly urban areas	I analyze trends	Done	High	3
32	Growth from 2018	Analyst	Compute growth % and km ²	I quantify expansion	Done	High	3
33	Change map 2018–2024	Analyst	Detect new urban pixels	I visualize expansion	Done	High	5

5.2.7 EPIC 7: Visualization and Dashboard

Objective: Develop an interactive web application for data exploration and result communication.

Table 11: EPIC 7: Visualization and Dashboard User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
34	Layer manager	User	Turn layers on/off	I explore results	Done	Medium	3
35	RGB & index layers	User	View NDVI, NDBI, BUI, MNDWI	I understand spectral behavior	Done	High	5
36	Urban mask layers	User	See urban per year	I compare years visually	Done	High	3
37	Growth map display	User	See 2018–2024 growth in red	I immediately see expansion zones	Done	High	3
38	Professional layout	User	Have clean UI panels	Dashboard looks academic	Done	Medium	5

5.2.8 EPIC 8: Export and Deliverables

Objective: Generate final outputs for reporting, archiving, and dissemination.

Table 12: EPIC 8: Export and Deliverables User Stories

ID	User Story	As a	I want to	So that	Status	Priority	Est. (SP)
39	Export CSV statistics	Analyst	Export area table	I use it in report	Done	High	2
40	Export classified maps	Analyst	Export 2018 & 2024 maps	I archive final products	Done	High	3
41	Export growth map	Analyst	Export change map	I use it in figures	Done	High	3

5.3 Project Release Planning and Execution

The project was executed through three sequential releases, each corresponding to a major phase of development and validation. This structured approach ensured incremental delivery of functionality while maintaining scientific rigor throughout the implementation process.

5.3.1 Release 1: Data and Classification Foundation

Duration: 6 Weeks (Sprint 1, Sprint 2, Sprint 3)

Objective: Establish the complete remote sensing processing pipeline, from raw data ingestion to initial land cover classification.

Included Features:

- Sentinel-2 data ingestion and cloud masking.
- Annual composite generation (2018–2024).
- Spectral index computation (NDVI, NDBI, MNDWI, BUI, DUI).
- Training dataset construction through polygon digitization.
- Balanced pixel sampling strategy implementation.
- Random Forest classifier training on 2018 reference data.

- Initial land cover classification for all study years (2018–2024).

Value Delivered: At the conclusion of Release 1, the project successfully produced scientifically valid land cover maps of the UAE for the entire study period. This was achieved through a reproducible and automated Google Earth Engine pipeline, forming the technical core of the research methodology. The release established the foundational workflow necessary for subsequent analytical phases.

5.3.2 Release 2: Validation, Correction, and Urban Growth Analytics

Duration: 6 Weeks (Sprint 4, Sprint 5, Sprint 6)

Objective: Enhance result reliability through systematic validation and enable quantitative urban growth analysis through accurate change detection.

Included Features:

- Confusion matrix computation and accuracy assessment.
- Calculation of validation metrics: Overall Accuracy, Producer’s Accuracy, User’s Accuracy, and Kappa coefficient.
- Implementation of post-classification correction rules (water and vegetation masking).
- Spatial smoothing through focal mode filtering.
- Robust urban area computation with `reduceRegion` bug resolution.
- Multi-year urban statistics table generation.
- Urban growth mapping (2018–2024 change detection).
- Export functionality for CSV statistics and classified raster products.

Value Delivered: Release 2 produced quantified, validated, and publishable urban growth statistics accompanied by comprehensive cartographic outputs. The systematic validation process ensured scientific credibility, while the bug fixes in area calculation guaranteed mathematical accuracy. This release constitutes the scientific results core of the thesis, providing both statistical evidence and spatial visualization of urbanization patterns.

5.3.3 Release 3: Dashboard, Visualization, and Final Delivery

Duration: 3 Weeks (Sprint 7, Sprint 8)

Objective: Transform analytical results into a professional decision-support system and prepare final deliverables for dissemination.

Included Features:

- Interactive Google Earth Engine dashboard interface.
- Layer selection system (RGB composites, spectral indices, classification results).
- Urban growth visualization with highlighted expansion areas.
- Integrated statistics panels and temporal charts.
- Year selector for temporal navigation.
- Professional legend implementation and layout optimization.
- Export functionality for publication-quality figures.
- Final map product generation.
- Report structuring and technical writing.

Value Delivered: The final release delivered a comprehensive visualization and communication platform that enables stakeholders to interactively explore urbanization patterns. The dashboard serves as both an analytical tool for experts and a communication medium for broader audiences. Combined with the finalized report structure and export products, this release ensures that research outcomes are accessible, reproducible, and suitable for academic and professional dissemination.

5.4 Release Management Strategy

The phased release approach provided several strategic advantages:

- Risk Mitigation: Technical challenges were identified and resolved early in the pipeline development phase.
- Incremental Validation: Each release produced verifiable outputs that could be assessed

before proceeding to subsequent phases.

- **Stakeholder Alignment:** Clear milestones facilitated communication with academic supervisors and ensured alignment with research objectives.
- **Quality Assurance:** The separation of classification development (Release 1), validation (Release 2), and visualization (Release 3) allowed focused attention on each aspect of scientific rigor.

This structured release planning ensured that the project maintained momentum while adhering to engineering standards of verification and validation at each development stage.

5.5 Project Management Outcomes

The Agile-Scrum approach facilitated the following outcomes:

- **Transparent Progress Tracking:** The product backlog provided a clear roadmap and allowed for continuous monitoring of completion status.
- **Risk Mitigation:** Technical challenges (e.g., the “0 km² bug” in area calculation) were identified early and addressed within targeted sprints.
- **Incremental Validation:** Each epic produced a verifiable output, ensuring that the scientific workflow remained robust and repeatable.
- **Stakeholder Alignment:** The structured backlog served as a communication tool, aligning technical implementation with research objectives.

This methodological rigor ensured that the project delivered scientifically valid, reproducible, and well-documented results suitable for engineering and geospatial analysis standards.

Chapter 6

6 Conclusion and Future Work

6.1 Overall Project Summary and Achievements

This project successfully designed and implemented a comprehensive, operational, and scientifically rigorous framework for monitoring and quantifying urban growth in the United Arab Emirates using satellite remote sensing and cloud-based geospatial processing. Beginning with raw Sentinel-2 imagery, a complete processing chain was developed encompassing image preprocessing, spectral feature engineering, supervised classification, accuracy assessment, post-classification refinement, spatial analysis, and interactive dashboard visualization.

The primary achievement of this work is the production of consistent, multi-temporal land cover maps spanning the period 2018–2024, with specific emphasis on urban area delineation. The system successfully automated the computation of annual urban extents, quantified year-over-year changes, and generated cumulative growth statistics at the national scale. Final results demonstrate a clear and continuous urban expansion trend in the UAE, with built-up area increasing from approximately 7,200 km² in 2018 to approximately 11,600 km² in 2024, representing an overall growth exceeding 60% over the six-year study period.

Beyond numerical results, a significant contribution of this project is the operational workflow implemented entirely within the Google Earth Engine platform. This ensures full reproducibility, scalability to large geographic regions, and the capacity to update analyses. The interactive dashboard further transforms technical outputs into an accessible decision-support and communication tool, enabling non-specialist users to visually explore maps, spectral indices, and statistical trends.

Methodologically, the project demonstrates that a carefully designed combination of spectral indices (NDVI, NDBI, MNDWI, BUI), Random Forest classification, and post-

classification logical and spatial rules can produce robust urban maps even in challenging environments such as the UAE, where desert, bare rock, and urban materials often exhibit similar spectral signatures. This project is also achieved with Agil Management Framework.

6.2 Answer to the Initial Problem Statement

The initial problem addressed by this project was the absence of a consistent, reproducible, and up-to-date national-scale monitoring framework for urban expansion in the UAE based on satellite data. Traditional statistics and isolated studies lack the temporal continuity, spatial detail, or automation required to support long-term planning and environmental assessment.

This project provides a concrete and operational solution through: - A fully automated multi-year urban mapping pipeline. - Quantitative and spatially explicit measurements of urban growth. - A validated classification approach with comprehensive accuracy assessment. - A visual analytics dashboard that makes results accessible and interpretable.

Thus, the project successfully fulfills its primary objective: transforming raw Earth observation data into reliable, decision-ready information about urbanization dynamics in the UAE.

6.3 Critical Discussion of Project Limitations

Despite strong results, several limitations must be acknowledged to maintain scientific rigor and contextualize findings.

First, the analysis relies primarily on Sentinel-2 imagery with 10–20 m spatial resolution. While sufficient for national-scale urban mapping, this resolution cannot detect very small buildings, narrow roads, or fine urban details. Mixed pixels, particularly at urban–desert or urban–mountain interfaces, remain unavoidable.

Second, although Random Forest is a robust and established classifier, the approach remains pixel-based. Spatial context, object shapes, and morphological patterns are only

partially considered, despite the post-classification smoothing and logical rules implemented.

Third, training data were manually digitized through visual interpretation. While extensive care was taken to ensure quality and representativeness, some degree of subjectivity and sampling bias may persist, particularly in spectrally ambiguous areas such as the rocky terrain in eastern UAE.

Finally, although a thorough accuracy assessment was conducted, validation remains limited by the availability of reliable reference data at the national scale and the absence of systematic field validation data.

These limitations do not invalidate the results but must be considered when interpreting absolute values and fine-scale spatial details.

6.4 Future Work and Perspectives

This project establishes several important directions for future improvement and scientific extension.

A first major improvement would involve integrating very high resolution imagery (e.g., PlanetScope, WorldView, or national aerial orthophotos) for selected years or regions. This would enable more precise delineation of urban structures and provide detailed validation of Sentinel-2 based results.

A second significant perspective is the adoption of Deep Learning methods, such as Convolutional Neural Networks or Transformer-based models, for semantic segmentation of urban areas. These methods can exploit spatial context and texture information more effectively than pixel-based classifiers and could substantially improve performance in complex landscapes such as mountainous and desert-urban transition zones.

Third, while the current analysis employs one composite per year, future work could transition to full time-series analysis using all available images. This would facilitate the study of intra-annual dynamics, detection of abrupt changes, and improved distinction between temporary and permanent land cover changes.

Another promising extension would involve integrating socio-economic and infrastructural data (population, road networks, industrial zones, planning boundaries) to progress from purely spatial mapping toward urban growth modeling and scenario simulation.

Finally, the dashboard could be expanded into a comprehensive decision-support platform, allowing users to query specific regions, generate custom reports, and interactively compare different urban growth indicators.

7 References

Recommended citation style: IEEE format

1. L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
2. N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, vol. 202, pp. 18–27, 2017.
3. H. Xu, "Modification of Normalised Difference Water Index (MNDWI) to enhance open water features in remotely sensed imagery," *International Journal of Remote Sensing*, vol. 27, no. 14, pp. 3025–3033, 2006.
4. Z. Zha, J. Gao, and S. Ni, "Use of normalized difference built-up index in automatically mapping urban areas from TM imagery," *International Journal of Remote Sensing*, vol. 24, no. 3, pp. 583–594, 2003.
5. J. W. Rouse, R. H. Haas, J. A. Schell, and D. W. Deering, "Monitoring vegetation systems in the Great Plains with ERTS," in *Third ERTS Symposium*, NASA SP-351, vol. 1, 1973, pp. 309–317.
6. European Space Agency, *Sentinel-2 User Handbook*, ESA Standard Document, 2015.
7. UN-Habitat, *Cities and Climate Change: Global Report on Human Settlements 2011*, Earthscan, 2011.
8. World Bank, *Urban Development Overview*, 2022. [Online]. Available: <https://www.worldbank.org/e>
9. J. R. Jensen, *Introductory Digital Image Processing: A Remote Sensing Perspective*, 4th ed. Pearson, 2015.

10. Q. Weng, "Remote sensing of impervious surfaces in the urban areas: Requirements, methods, and trends," *Remote Sensing of Environment*, vol. 117, pp. 34–49, 2012.

Appendix A: Google Earth Engine Code

A Main Processing Script

```
// Excerpt from main GEE script for Sentinel-2 processing and classification
var s2 = ee.ImageCollection('COPERNICUS/S2_SR')
    .filterBounds(UAE)
    .filterDate('2018-01-01', '2024-12-31')
    .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20));

// Cloud masking function
function maskS2clouds(image) {
    var scl = image.select('SCL');
    var cloudMask = scl.neq(9); // Remove clouds
    return image.updateMask(cloudMask);
}
```

B Classification Function

```
// Random Forest classifier training and application
var training = urban.merge(vegetation).merge(desert).merge(water);
var bands = ['B2', 'B3', 'B4', 'B8', 'NDVI', 'NDBI', 'MNDWI', 'BUI'];

var classifier = ee.Classifier.smileRandomForest(100)
    .train(training, 'class', bands);

var classified = composite.select(bands).classify(classifier);
```

Appendix B: Detailed Accuracy Assessment Tables

Table 13: Complete Confusion Matrix for 2018 Classification

Reference/Predicted	Urban	Vegetation	Bare Soil	Water	Total
Urban	165	8	15	2	190
Vegetation	6	142	20	0	168
Bare Soil	12	18	205	5	240
Water	0	1	3	96	100
Total	183	169	243	103	698

Table 14: Accuracy Metrics Summary (2018–2024 Average)

Class	Producer's Accuracy (%)	User's Accuracy (%)
Urban	86.8	87.3
Vegetation	84.5	84.5
Bare Soil	85.4	85.8
Water	96.0	96.0

Appendix C: Training and Validation Sample Locations

Table 15: Training and Validation Sample Statistics

Land Cover Class	Training Samples	Validation Samples
Urban	785	336
Vegetation	450	193
Bare Soil	812	348
Water	214	92
Total	2,261	969